Table 1. Metadata for acoustical monitoring sites at Gates of the Arctic National Park and Preserve (GAAR). Dates reflect each site's recorded NVSPL deployment period(s).

Site	Site Name	Dates	Vegetation	Wilderness	Elevation (m)	Latitude	Longitude
GAARAMLA	NA		NA	NA	NA	NA	NA
GAARANAK	NA		NA	NA	NA	NA	NA
GAARARRI	NA		NA	NA	NA	NA	NA
GAARFLOR	NA		NA	NA	NA	NA	NA
GAARHAMM	NA		NA	NA	NA	NA	NA
GAARITKI	NA		NA	NA	NA	NA	NA
GAARKURU	NA		NA	NA	NA	NA	NA
GAARMWAL	NA		NA	NA	NA	NA	NA
GAARNARV	NA		NA	NA	NA	NA	NA
GAARNNUT	NA		NA	NA	NA	NA	NA
GAARNUTU	NA		NA	NA	NA	NA	NA
GAARNWAL	NA		NA	NA	NA	NA	NA
GAARPALA	NA		NA	NA	NA	NA	NA
GAARSELB	NA		NA	NA	NA	NA	NA
GAARUPNO	NA		NA	NA	NA	NA	NA
GAARWALK	NA		NA	NA	NA	NA	NA

Table 2. Summary of sound levels at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2018. For each site, includes the 90th percentile (LA10), existing (LA50), natural (LAnat), and 10th percentile (LA90) ambient sound levels (dB re 20 uPa, A−weighted broadband, 12.5−20,000 Hz) during the day (0700−1900) and night (1900−0700).

Time	SiteID	LA10	LA50	LAnat	LA90
Day	GAARARRI	44.4	41.3	41.3	38.6
Night	GAARARRI	49.2	47.3	47.3	45.0
Day	GAARFLOR	32.5	25.3	24.9	20.9
Night	GAARFLOR	32.0	24.4	24.4	21.5
Day	GAARUPNO	31.2	23.2	23.0	19.7
Night	GAARUPNO	30.1	24.1	24.1	21.7

Table 3. Number of days of audio analyzed for the Listening Center and SPLAT detection methods at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2018 − the basis for the analysis−period text reported in the Methods section.

Type	DaysText
ListeningCenter	
SPLAT	29−76

Table 4. Executive summary for Gates of the Arctic National Park and Preserve (GAAR), Summer 2018: mean overall noise audibility, top contributing noise sources, and ambient sound levels (LA50/LAnat, day and night) by site.

SiteID	Noise	Aircraft	Watercraft	NA 4	LA50 Day	LAnat Day	LA50 Night	LAnat Night
GAARARRI	0.4	0.4	NA	NA	41.3	41.3	47.3	47.3
GAARFLOR	5.0	5.0	0	NA	25.3	24.9	24.4	24.4
GAARUPNO	1.2	1.2	NA	NA	23.2	23.0	24.1	24.1

Table 5. Noise impact (dB, existing minus natural ambient) and Listening Area Reduction (%) by site at Gates of the Arctic National Park and Preserve (GAAR), Summer 2018, for day, night, and the combined 24−hour period.

SiteID	DayImpact	DayLAR	NightImpact	NightLAR	AllImpact	AllLAR
GAARARRI	0.0	0.0	0	0	0.0	0.0
GAARFLOR	0.4	8.8	0	0	0.2	4.4
GAARUPNO	0.2	4.5	0	0	0.1	2.3

Table 6. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2018.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARARRI	0.0	0	No Sound Audible	0.0	No Sound Audible	0.0	0.2	0.2	0.0	0.0	0.0
GAARARRI	1.1	1	Aircraft	1.1	Jet	0.4	0.1	0.5	112.8	109.1	111.9
GAARARRI	1.2	1	Aircraft	1.2	Prop	1.4	0.1	1.5	176.9	137.8	173.9

Table 7. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2018.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARFLOR	1.1	1	Aircraft	1.1	Jet	3.8	1.2	5.1	253.6	277.6	259.4
GAARFLOR	1.1	1	Aircraft	1.1	Jet, Military	0.0	0.2	0.2	0.0	95.6	95.6
GAARFLOR	1.2	1	Aircraft	1.2	Prop	7.0	0.7	7.7	390.6	304.7	382.8
GAARFLOR	1.3	1	Aircraft	1.3	Helicopter	0.1	0.0	0.1	548.0	0.0	548.0
GAARFLOR	3.0	3	Watercraft	3.0	Watercraft	0.1	0.0	0.1	252.8	0.0	252.8

Table 8. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2018.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARUPNO	1.1	1	Aircraft	1.1	Jet	1.4	1.2	2.6	169.2	240.1	202.4
GAARUPNO	1.2	1	Aircraft	1.2	Prop	1.4	0.1	1.5	329.8	230.8	327.2

Table 9. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2018.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	3600	3600	3600	3600	3600	3600	3600	3600	2881	3151	481	1150	871	472	1604	521	1738	2241	3242	3187	2835	3600	3600	3600	GAARARRI
50%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARARRI
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARARRI
Mean	3173	3469	3600	3206	3600	3600	3600	3114	3065	2578	1922	2733	2472	2218	2791	2279	2870	2780	3021	3298	2924	3509	3519	3600	GAARARRI

Table 10. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2018.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	2274	1492	462	1048	416	718	2738	698	107	198	162	232	318	212	193	140	134	180	453	348	1100	1495	1623	1076	GAARFLOR
50%	3600	3600	3600	3600	3600	3600	3600	3600	2940	2069	1117	1188	1404	1986	916	773	1822	2423	3600	3600	3600	3600	3600	3600	GAARFLOR
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3306	3600	3048	3410	3600	3600	3600	3600	3600	3600	3600	3600	GAARFLOR
Mean	3109	2993	2434	2485	2198	2564	3114	2164	1387	1294	931	978	1110	1328	848	738	1195	1480	2226	2143	2785	3044	3043	2498	GAARFLOR

Table 11. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2018.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	3600	3600	3600	843	555	359	3234	3246	1706	729	696	688	1845	492	577	454	764	2657	401	833	3392	3600	3363	3600	GAARUPNO
50%	3600	3600	3600	3600	3600	3174	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	2241	3600	3600	3600	3600	3600	GAARUPNO
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARUPNO
Mean	3283	3600	3600	2701	2253	1845	3227	3263	3196	2532	2270	2292	2903	2036	2275	1980	2490	3156	1701	2389	3173	3600	3285	3469	GAARUPNO

Table 12. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2018.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARARRI	No Sound Audible	0.0	0	0	0	0	0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0	0
GAARARRI	Jet	0.1	0	0	0	0	0	0	0.1	0.2	0.1	0.1	0.1	0.1	0.2	0.1	0.2	0.1	0.0	0.0	0.1	0.0	0	0	0
GAARARRI	Prop	0.0	0	0	0	0	0	0	0.3	0.0	0.4	1.5	0.7	1.1	1.0	0.3	0.6	0.4	0.4	0.2	0.1	0.3	0	0	0

Table 13. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2018.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARFLOR	Jet	0.0	0.4	1.3	2.0	1.9	1.3	0.7	1.1	0.5	2.0	0.5	0.8	7.2	2.1	4.3	4.6	3.0	0.7	0.3	0.9	0.2	0.5	0.4	0.0
GAARFLOR	Jet, Military	0.0	0.0	0.1	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
GAARFLOR	Prop	0.7	0.3	0.0	0.0	0.3	0.0	0.0	1.0	4.5	3.4	12.6	11.3	4.4	3.5	11.8	8.3	5.9	6.6	2.2	1.2	0.7	0.3	0.6	1.8
GAARFLOR	Helicopter	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.2	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
GAARFLOR	Watercraft	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.0	0.0	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Table 14. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2018.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARUPNO	Jet	0.2	0	0	1.1	2.4	2.7	0.5	0.3	0.3	0.6	0.4	0	0.2	0.2	0.8	0.9	0.6	0.2	2.3	0.8	0.1	0	0.2	0.2
GAARUPNO	Prop	0.0	0	0	0.0	0.0	0.0	0.0	0.0	0.1	0.9	3.6	3	0.2	2.2	0.7	1.1	0.4	0.2	0.6	0.5	0.2	0	0.0	0.0

Table 15. Percent time above sound level thresholds for acoustic records collected at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2018. Values show the percent of time ANS−weighted sound levels (20−1,250 Hz) and the full measured frequency range (12.5−20,000 Hz) exceeded 35, 45, 52, and 60 dB (LAeq,1s), by site and day/night.

Site ID	Frequency (Hz)	35 dB Day	45 dB Day	52 dB Day	60 dB Day	35 dB Night	45 dB Night	52 dB Night	60 dB Night
GAARARRI	Full (12.5−20,000)	100.0	18.4	0.1	0	100.0	80.2	0.6	0
GAARARRI	ANS (20−1,250)	99.3	9.4	0.0	0	100.0	72.1	0.1	0
GAARFLOR	Full (12.5−20,000)	9.7	0.3	0.0	0	10.8	0.2	0.0	0
GAARFLOR	ANS (20−1,250)	4.4	0.1	0.0	0	3.7	0.0	0.0	0
GAARUPNO	Full (12.5−20,000)	6.2	1.2	0.2	0	3.9	0.5	0.1	0
GAARUPNO	ANS (20−1,250)	0.3	0.0	0.0	0	0.2	0.0	0.0	0

Figure 1. GAARARRI: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2018.

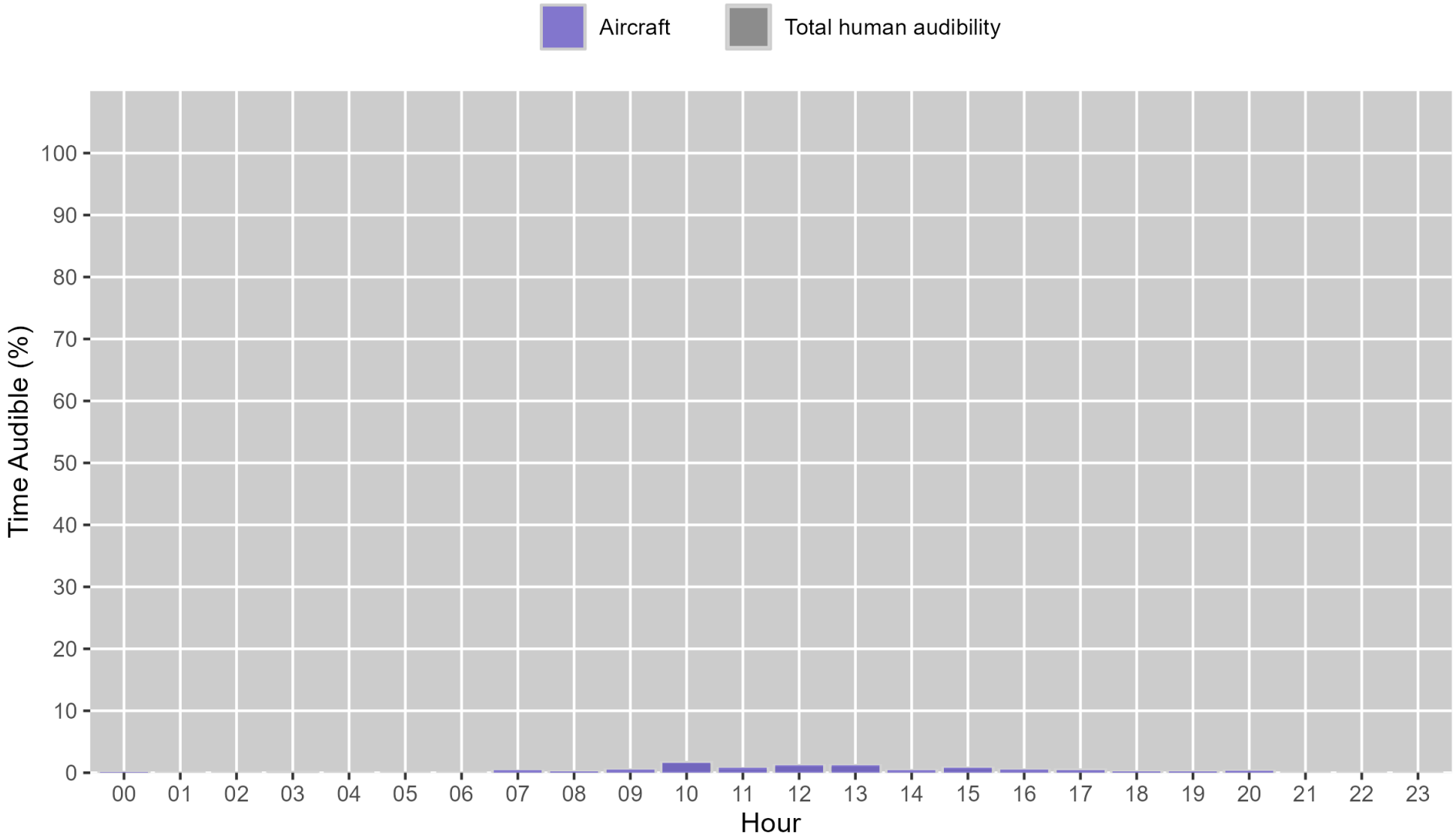


Figure 2. GAARARRI: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2018.

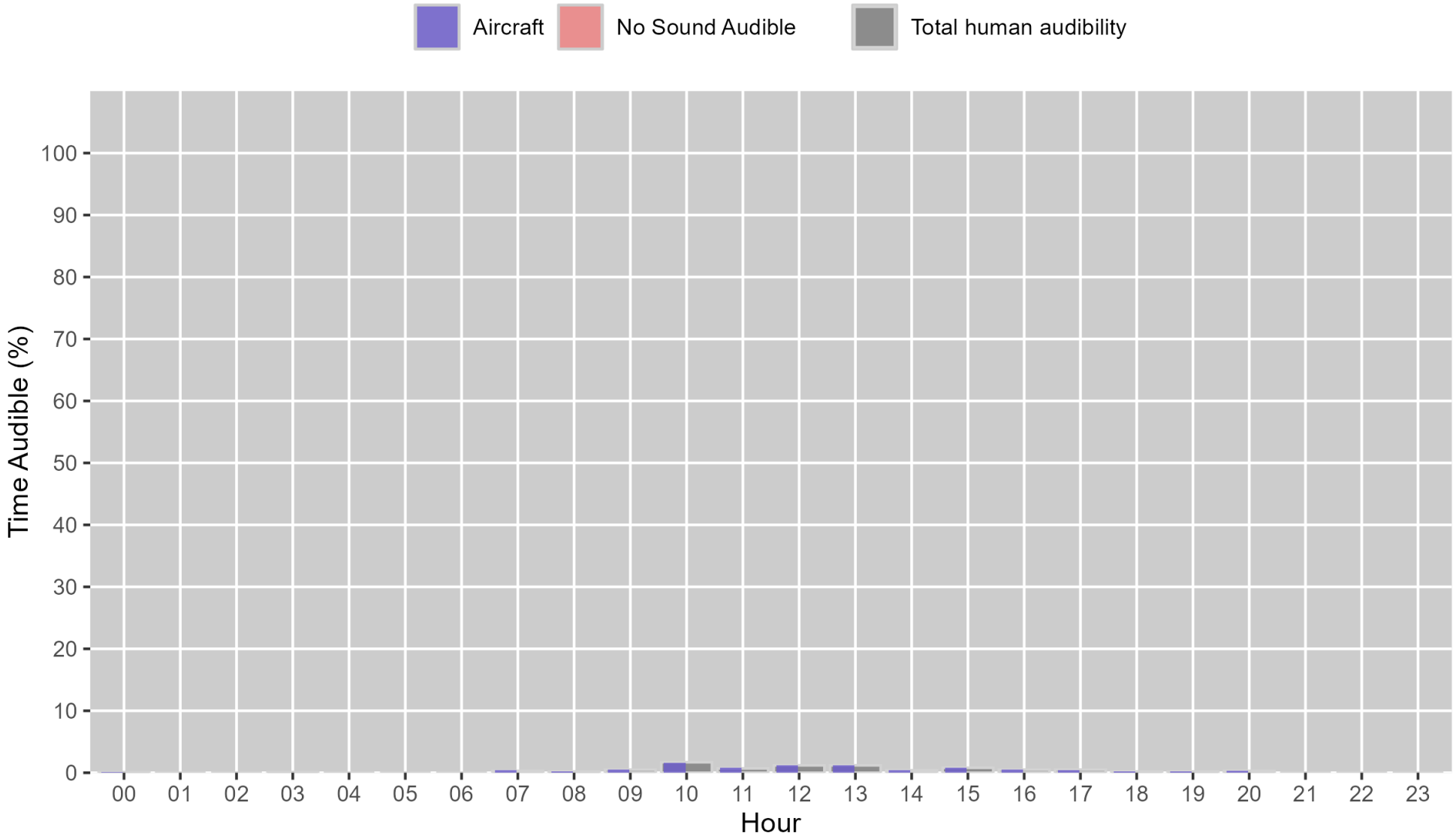


Figure 3. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARARRI in Summer, 2018, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

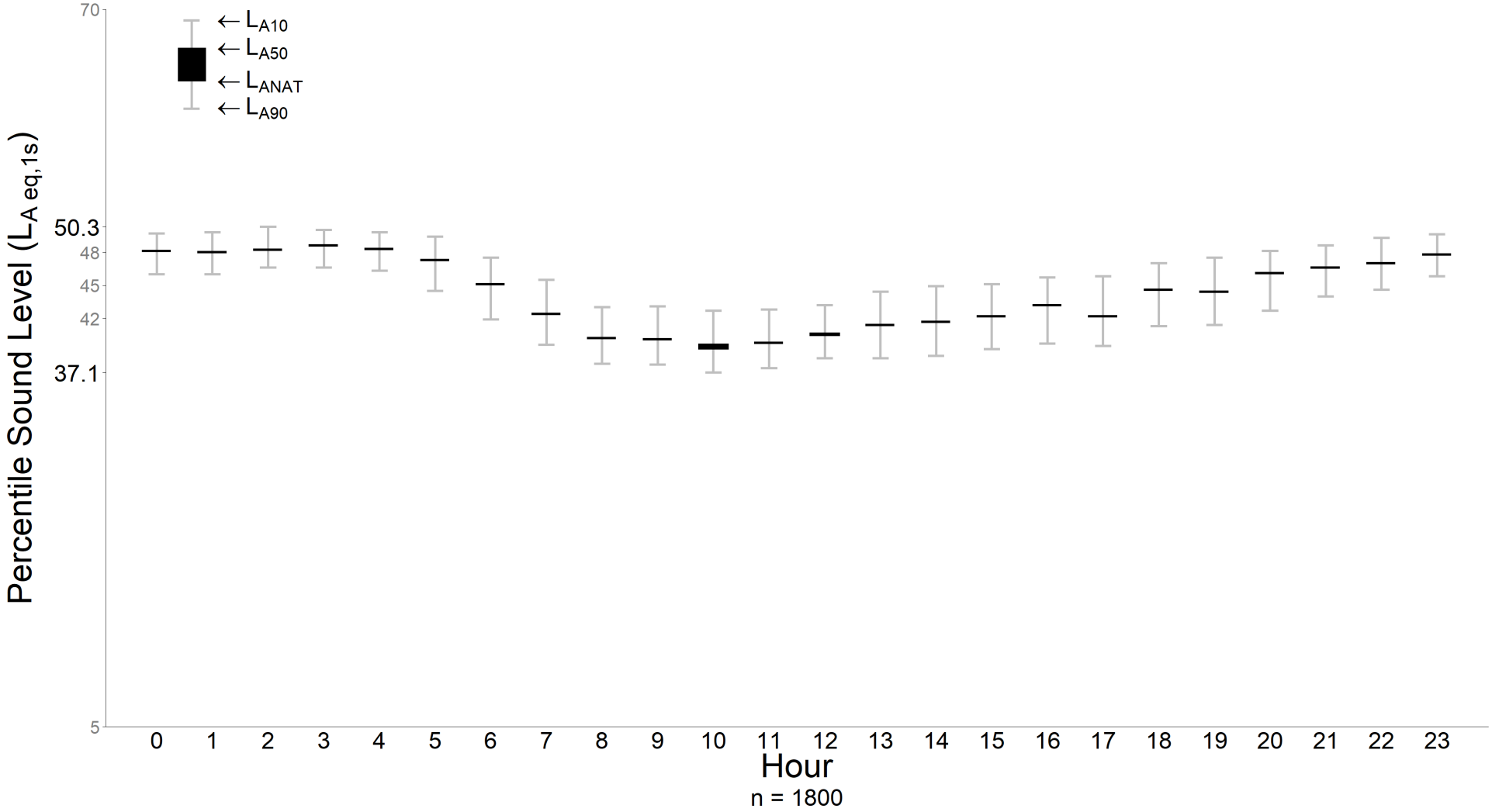


Figure 4. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARARRI in Summer, 2018.

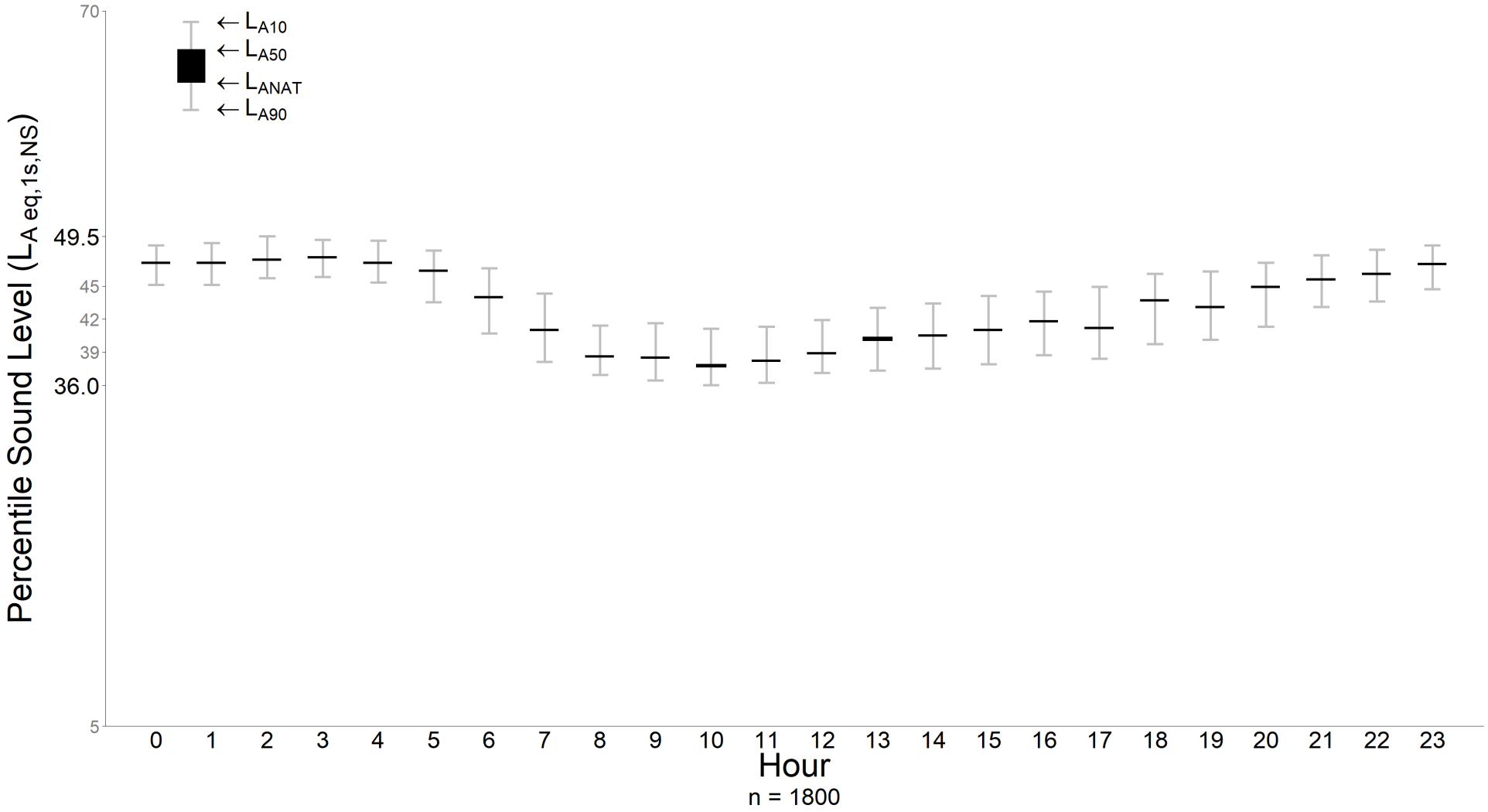


Figure 5. GAARARRI: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

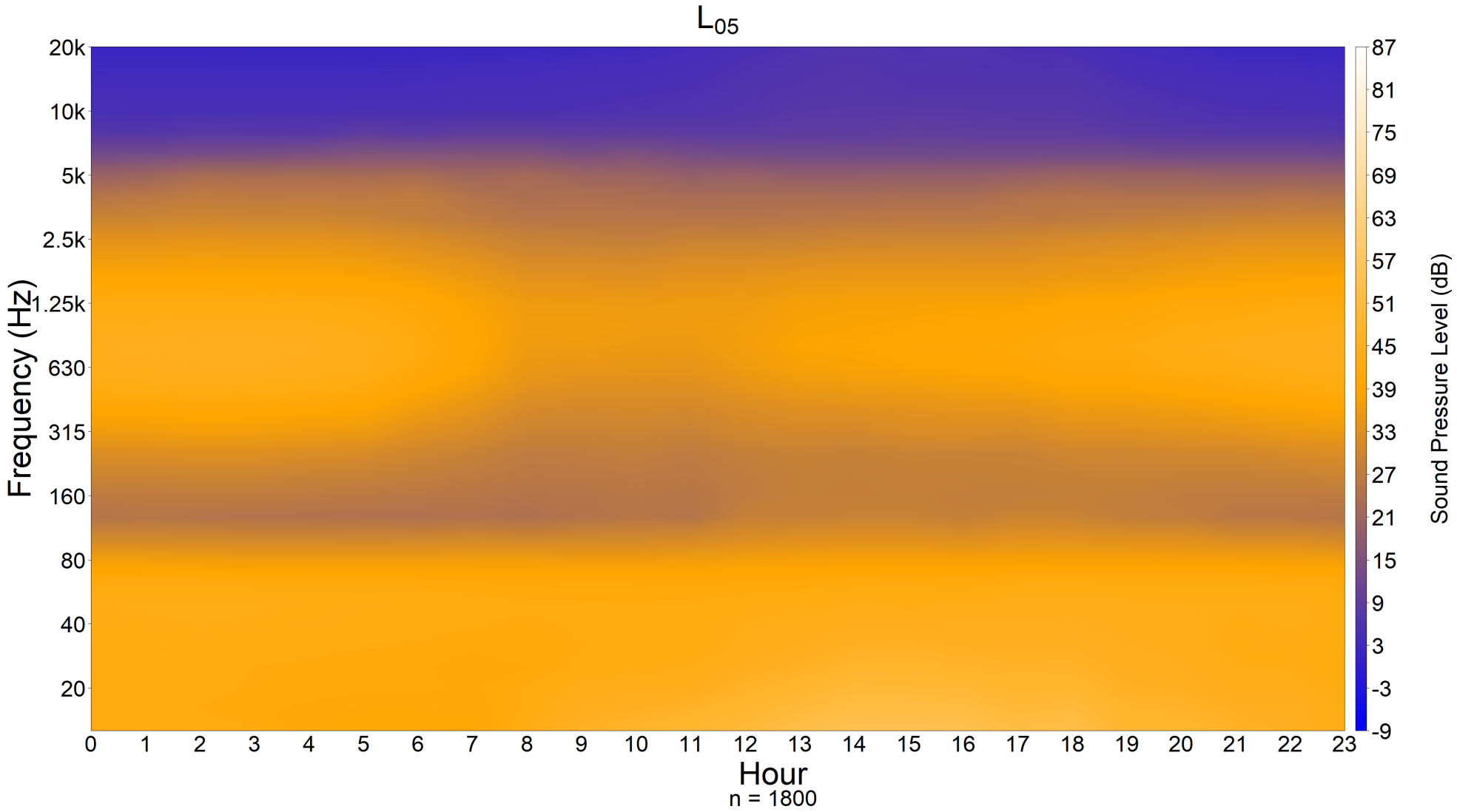


Figure 6. GAARARRI: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

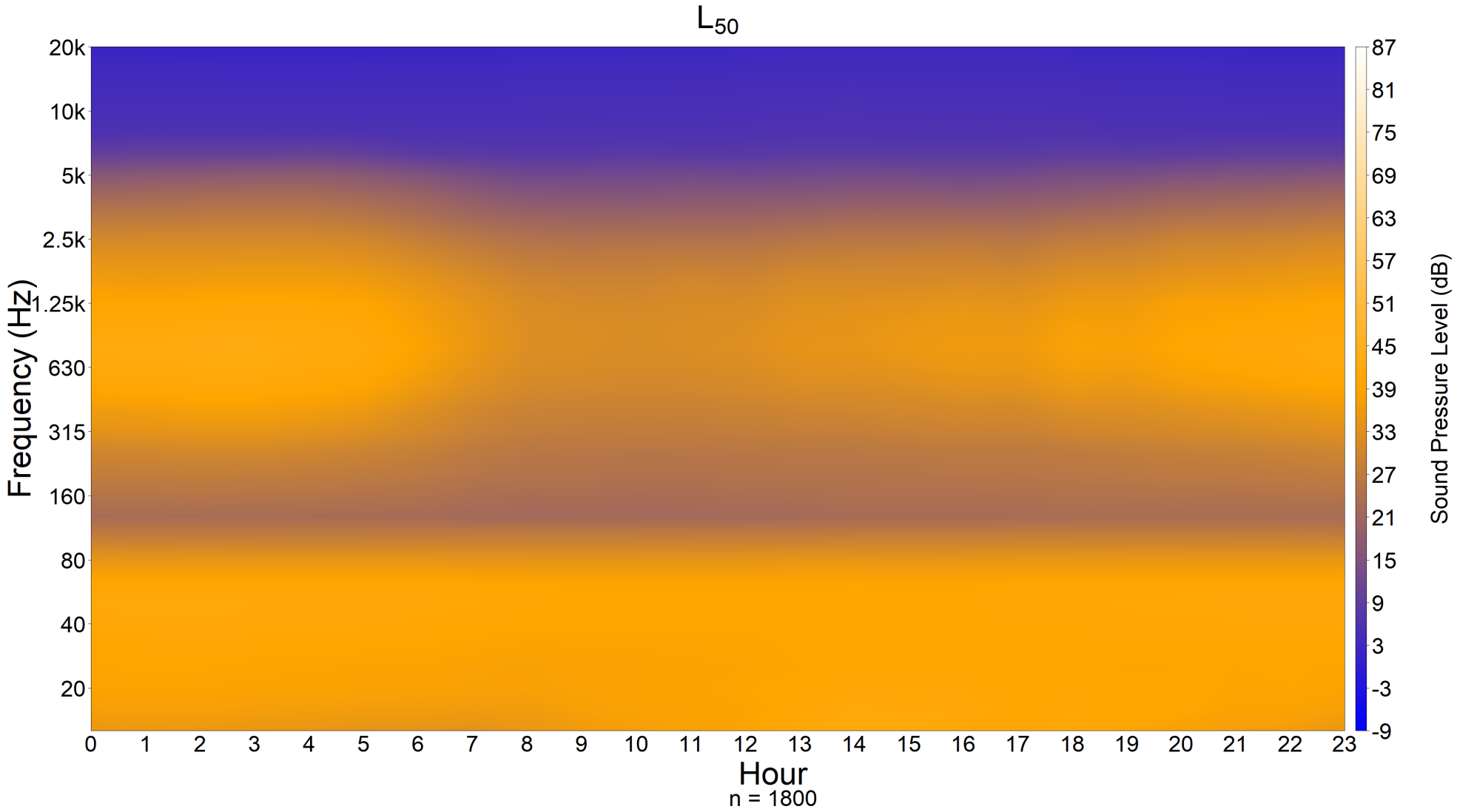


Figure 7. GAARARRI: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

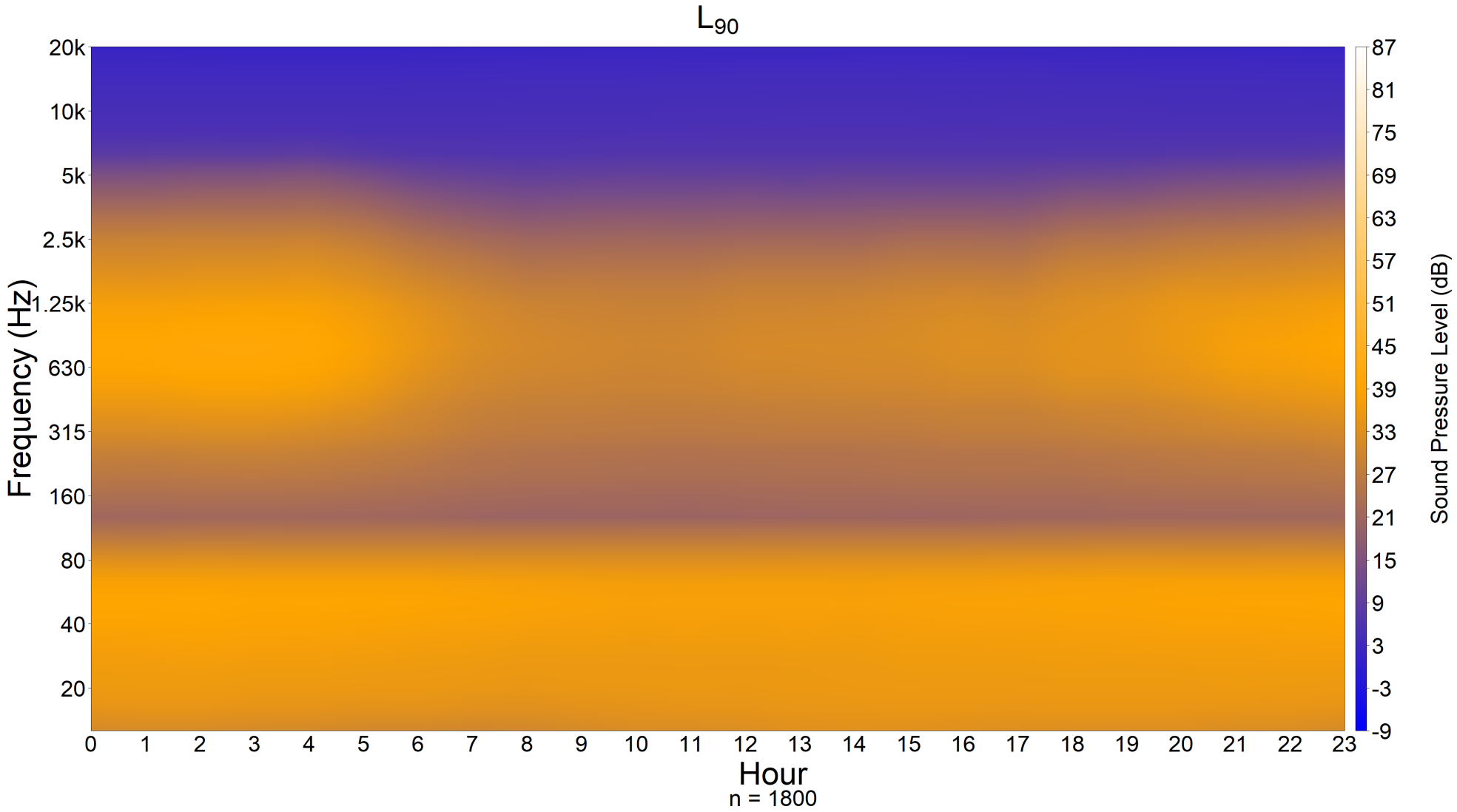


Figure 8. GAARARRI: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

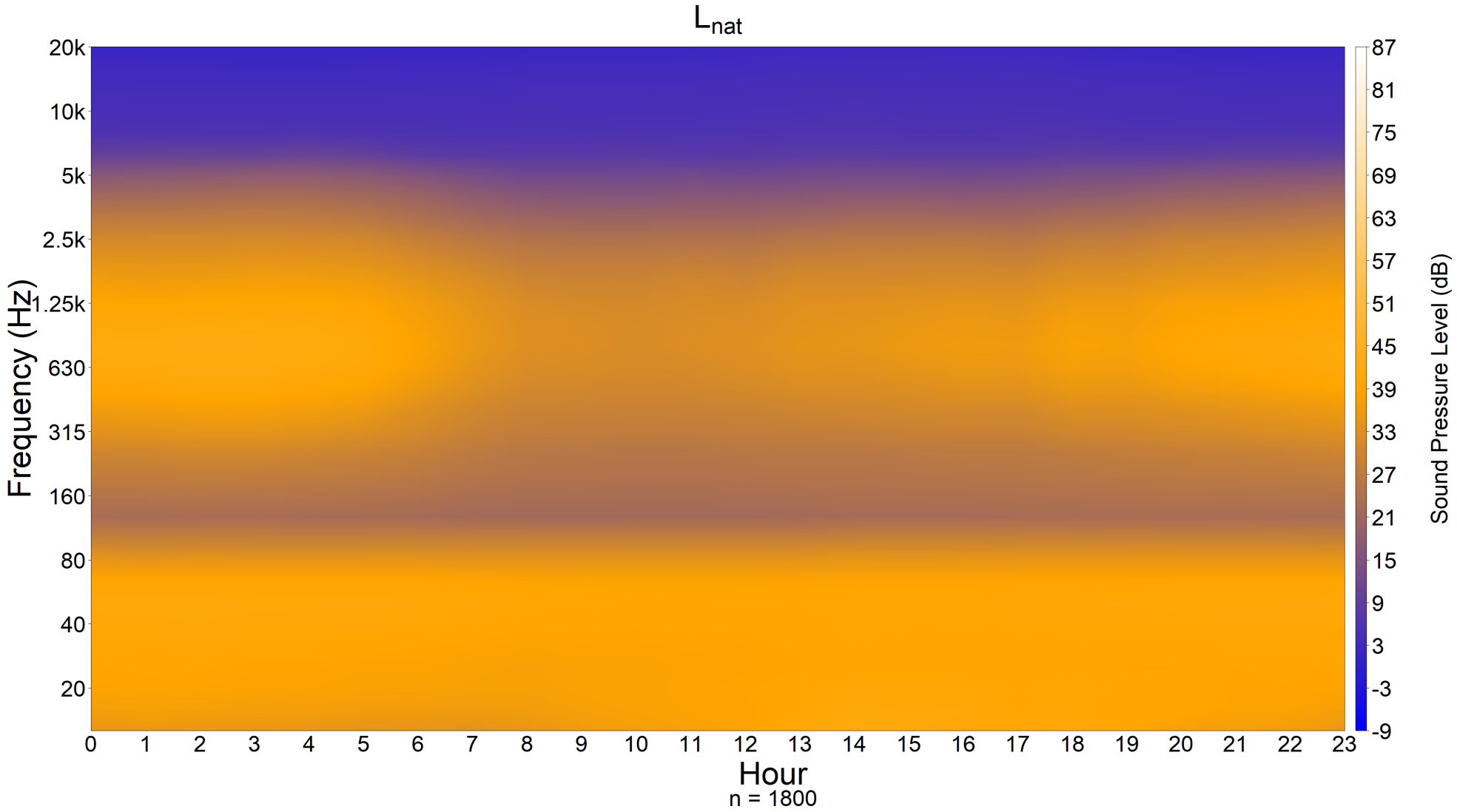


Figure 9. GAARARRI: median and mean duration of noise−free intervals by hour of day, in Summer 2018.

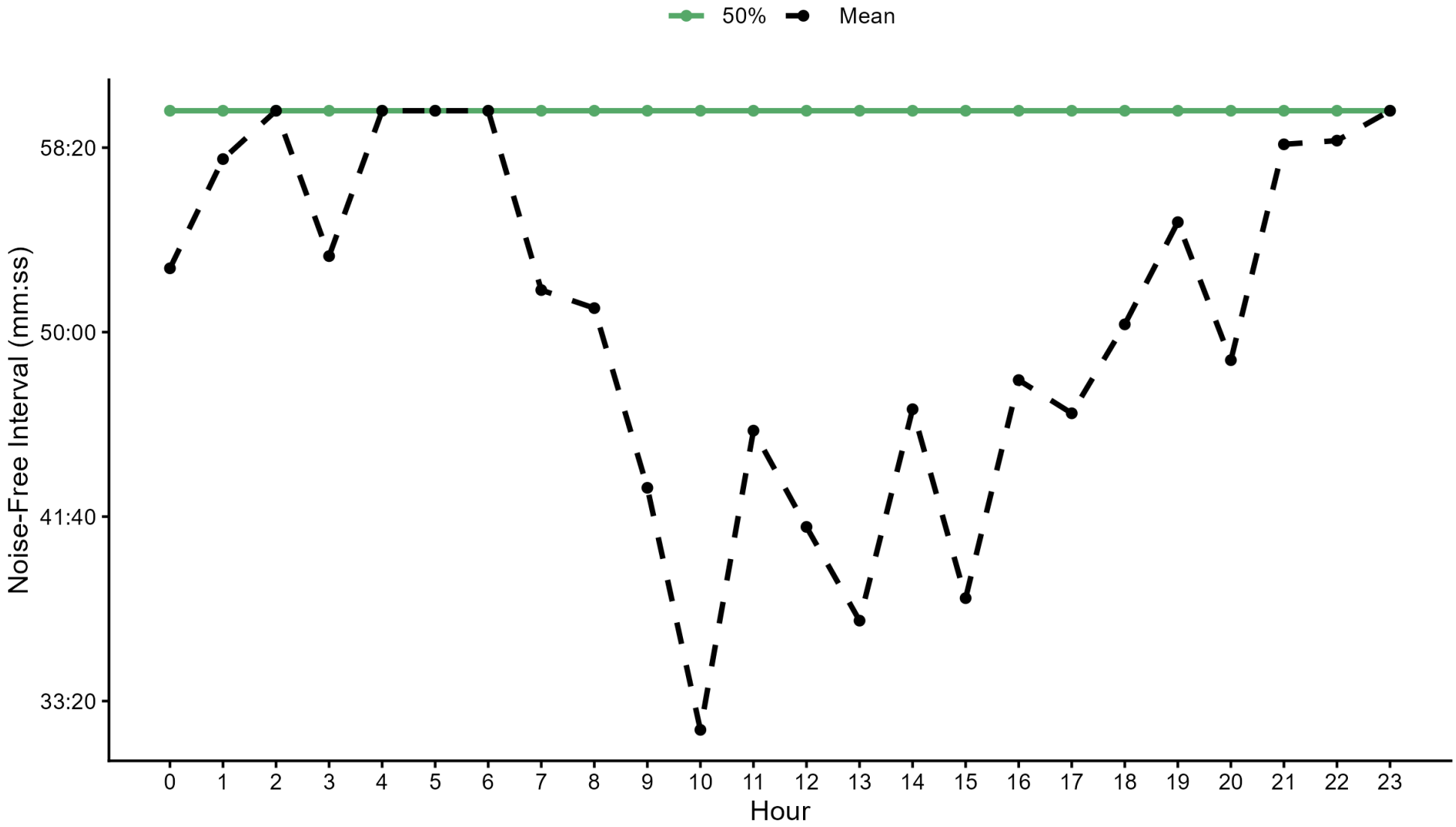


Figure 10. GAARARRI: percent of time audible by sound source category (noise/natural), in Summer 2018.

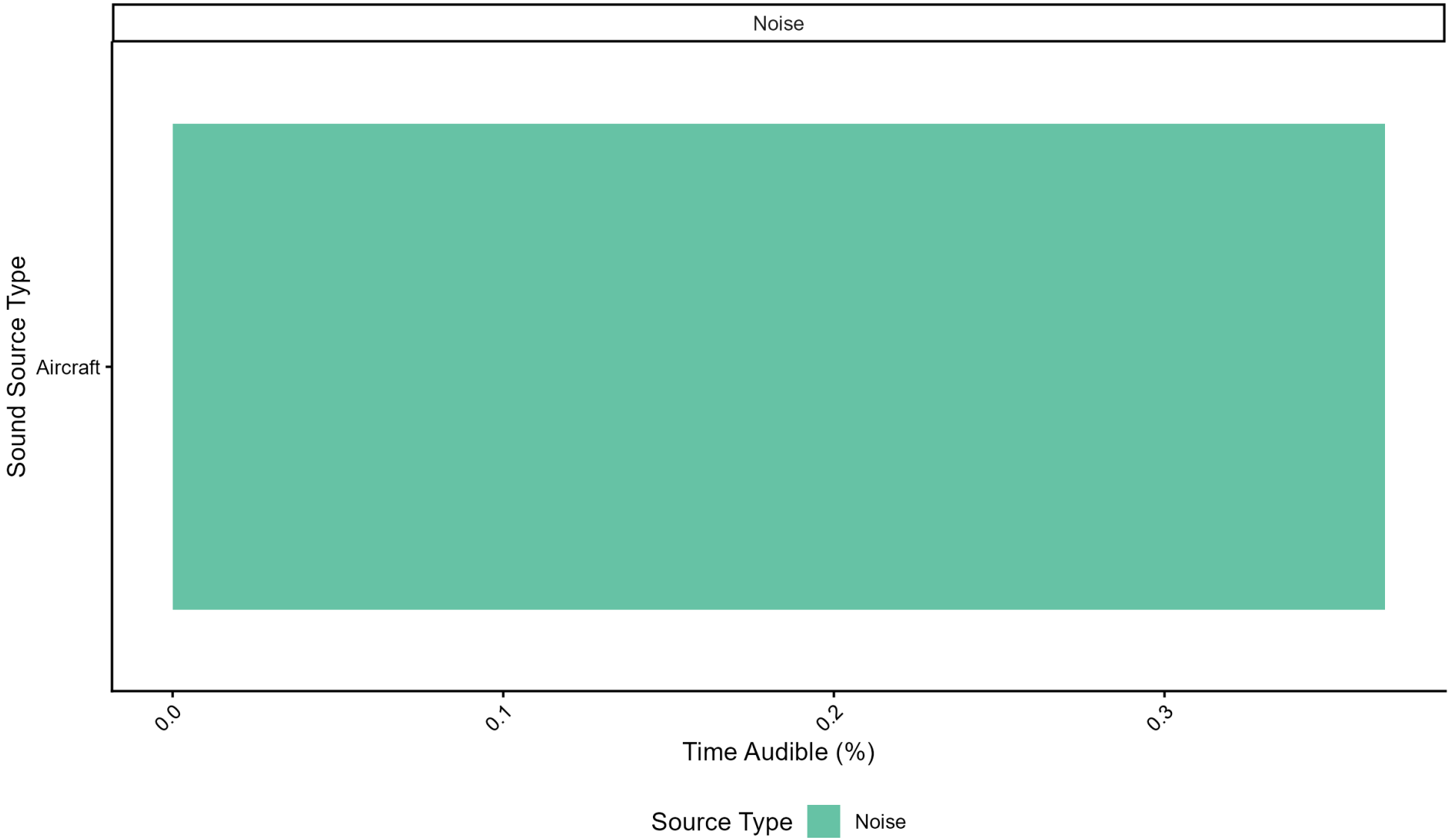


Figure 11. GAARARRI: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2018.

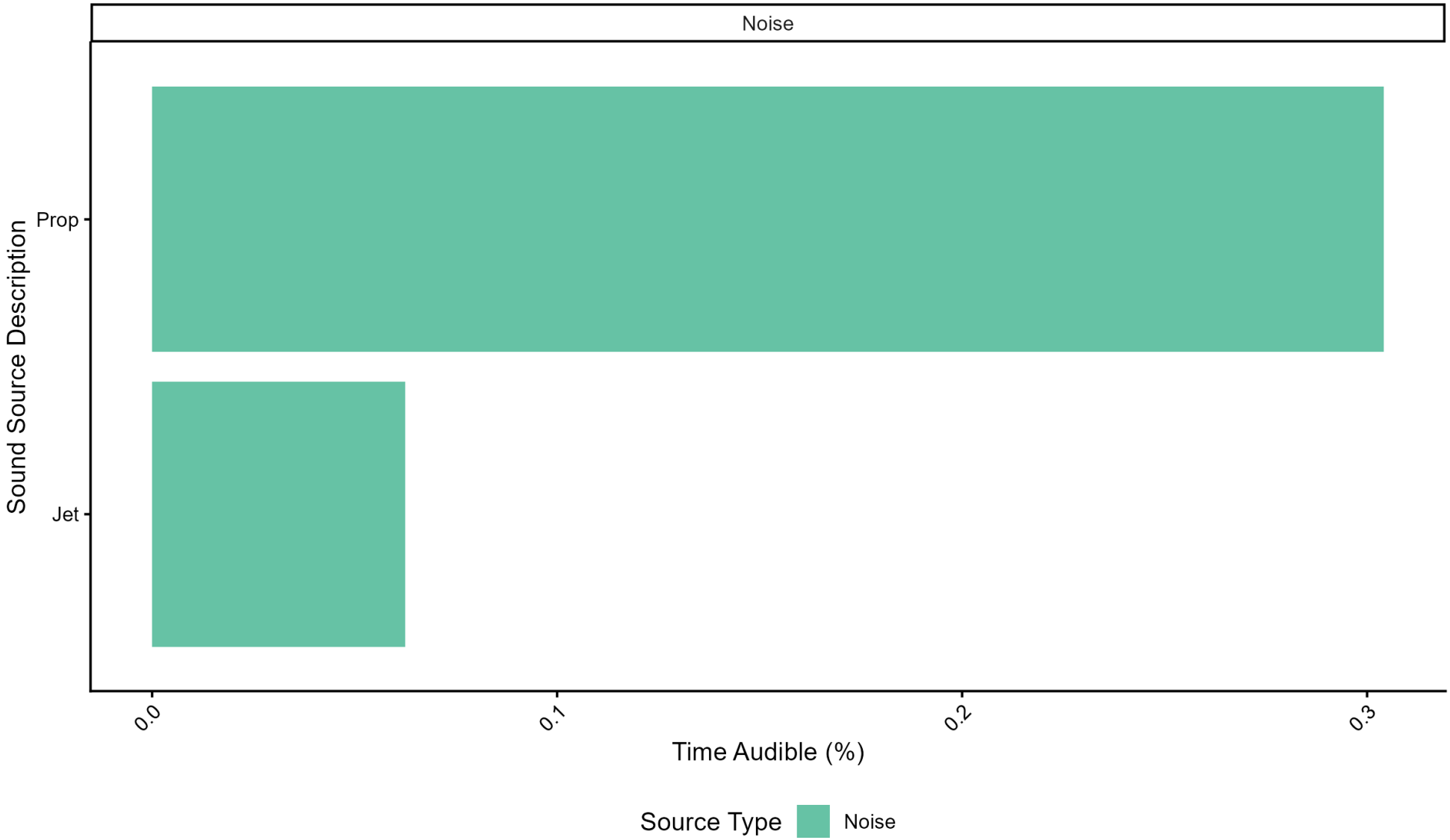


Figure 12. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARARRI in Summer, 2018. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

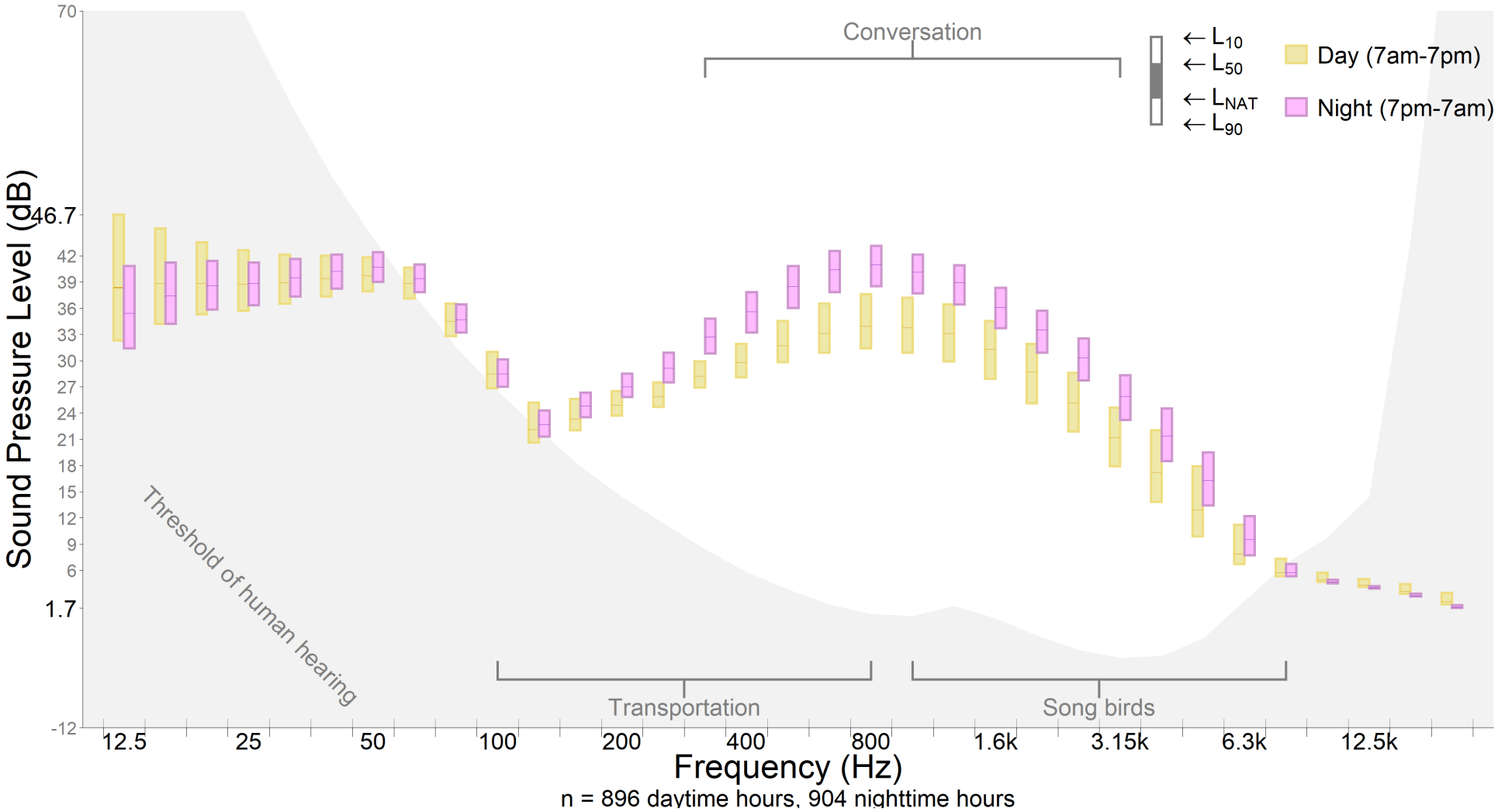


Figure 13. GAARFLOR: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2018.

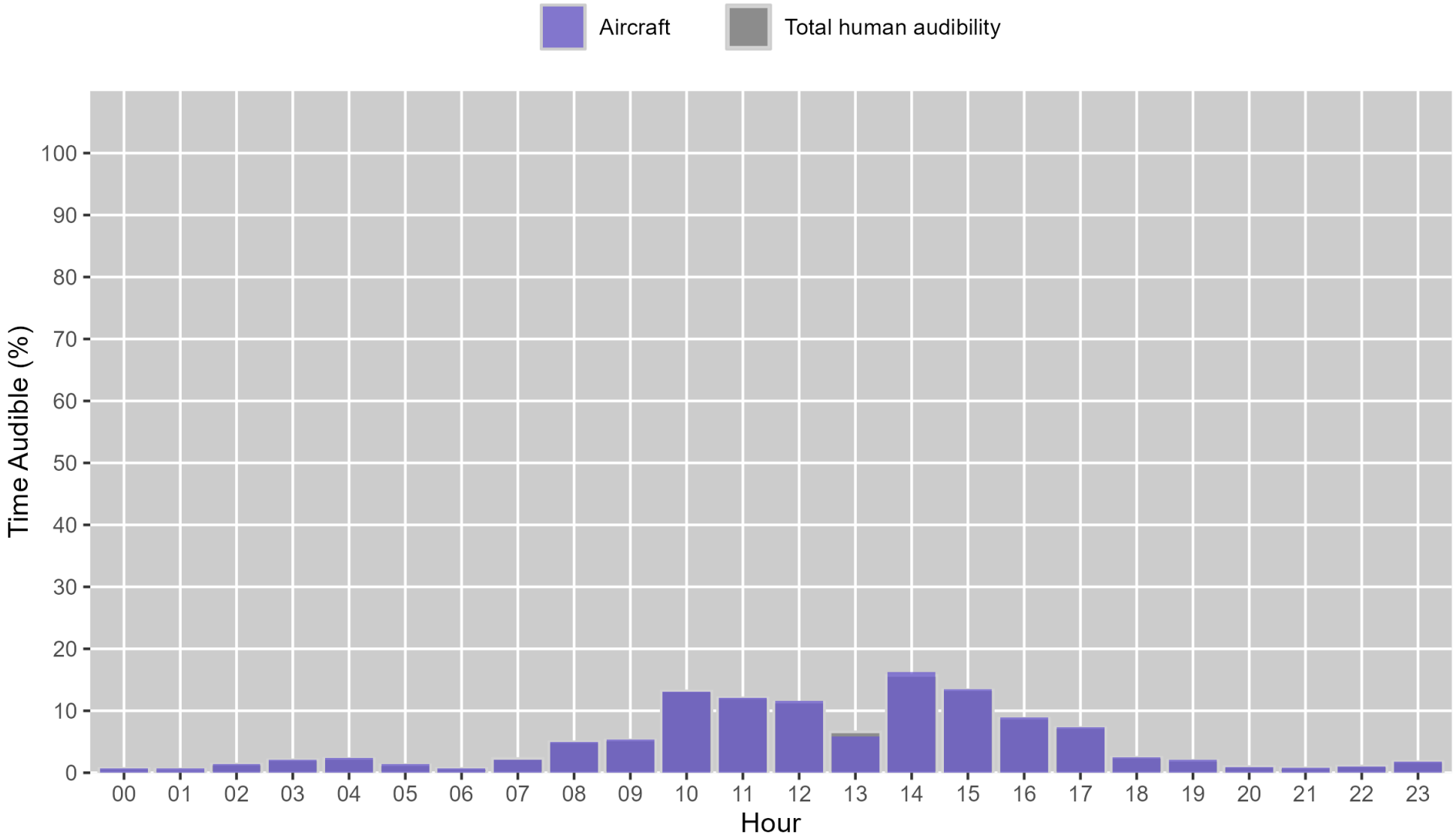


Figure 14. GAARFLOR: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2018.

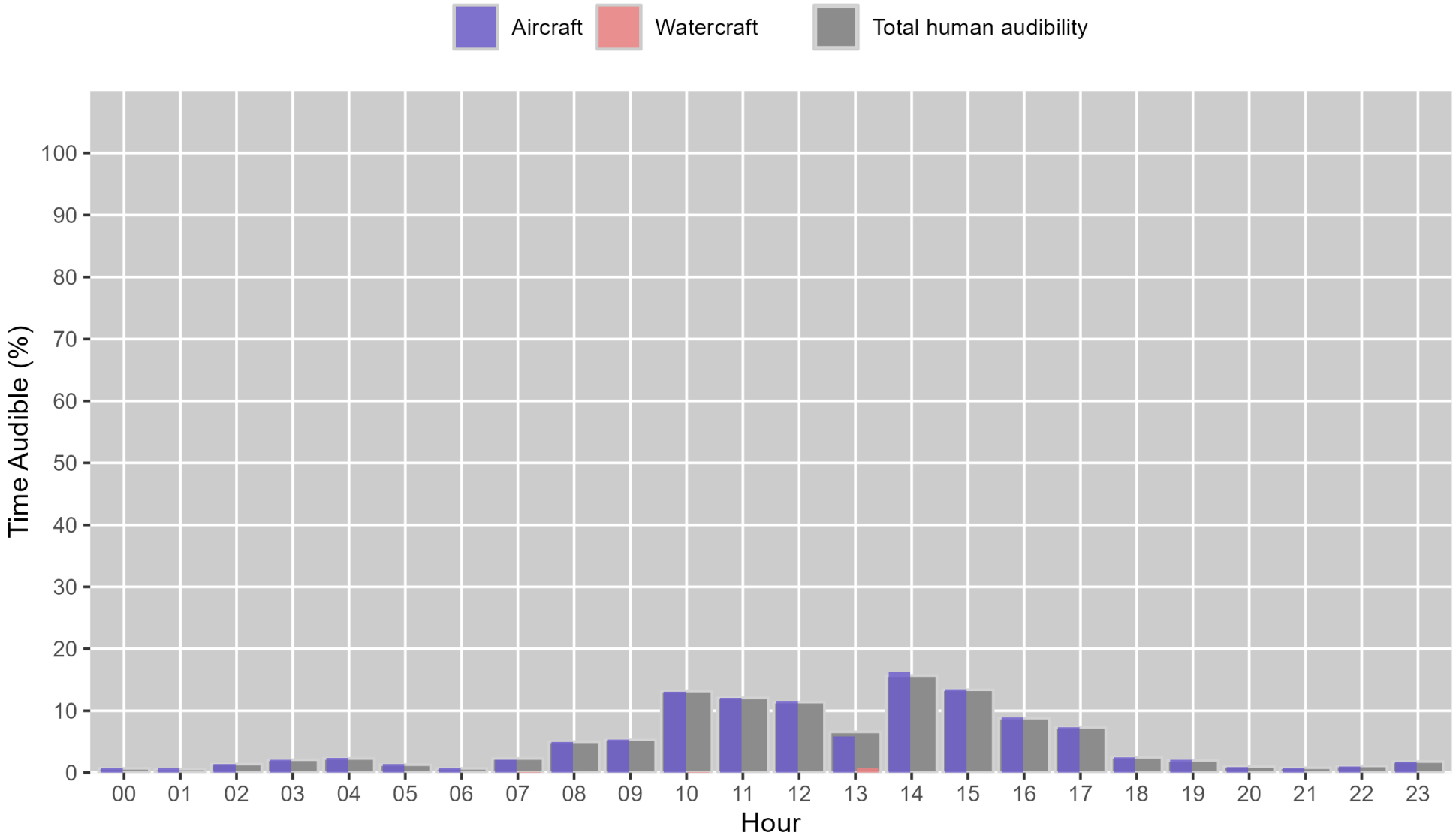


Figure 15. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARFLOR in Summer, 2018, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

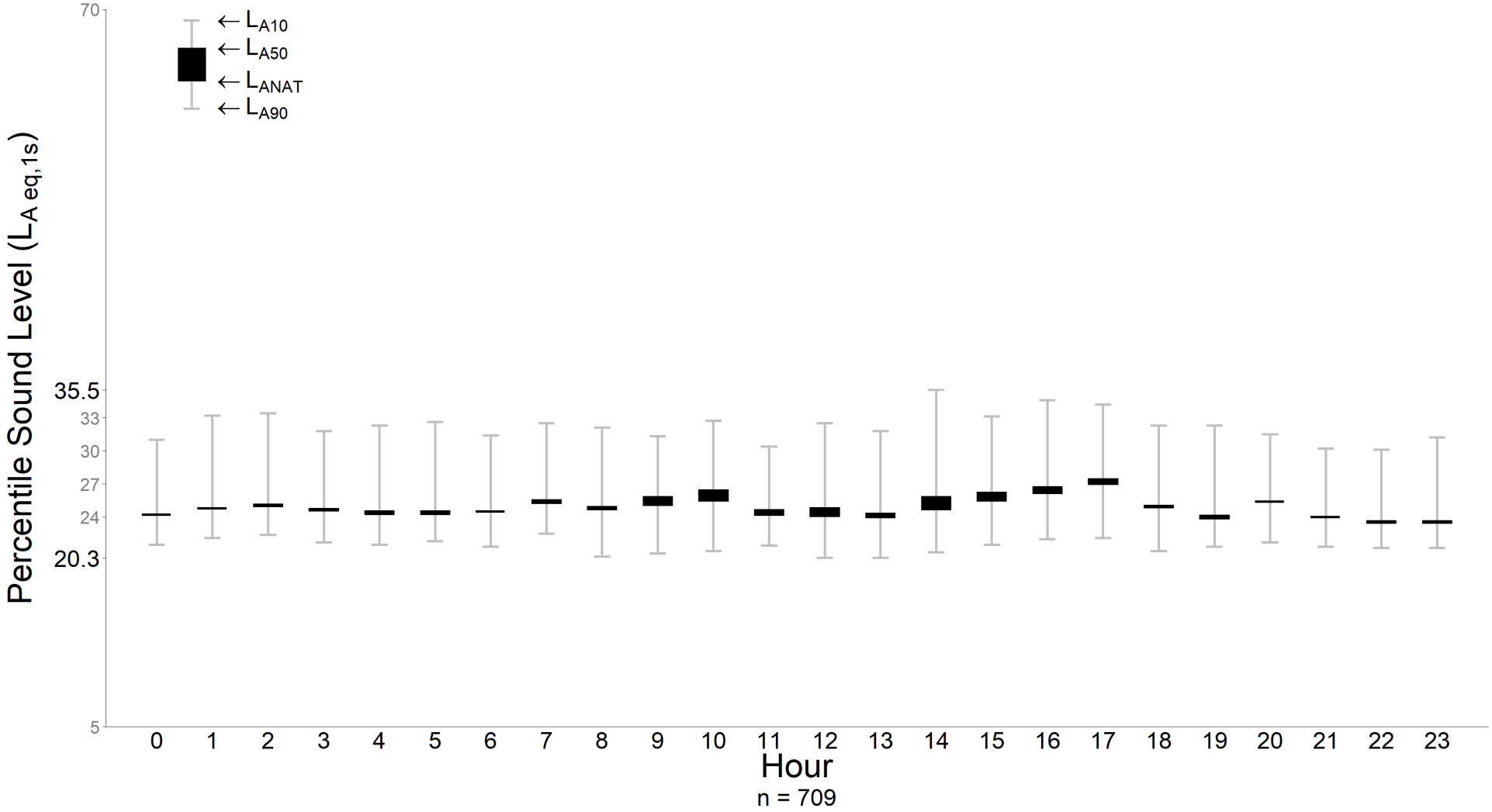


Figure 16. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARFLOR in Summer, 2018.

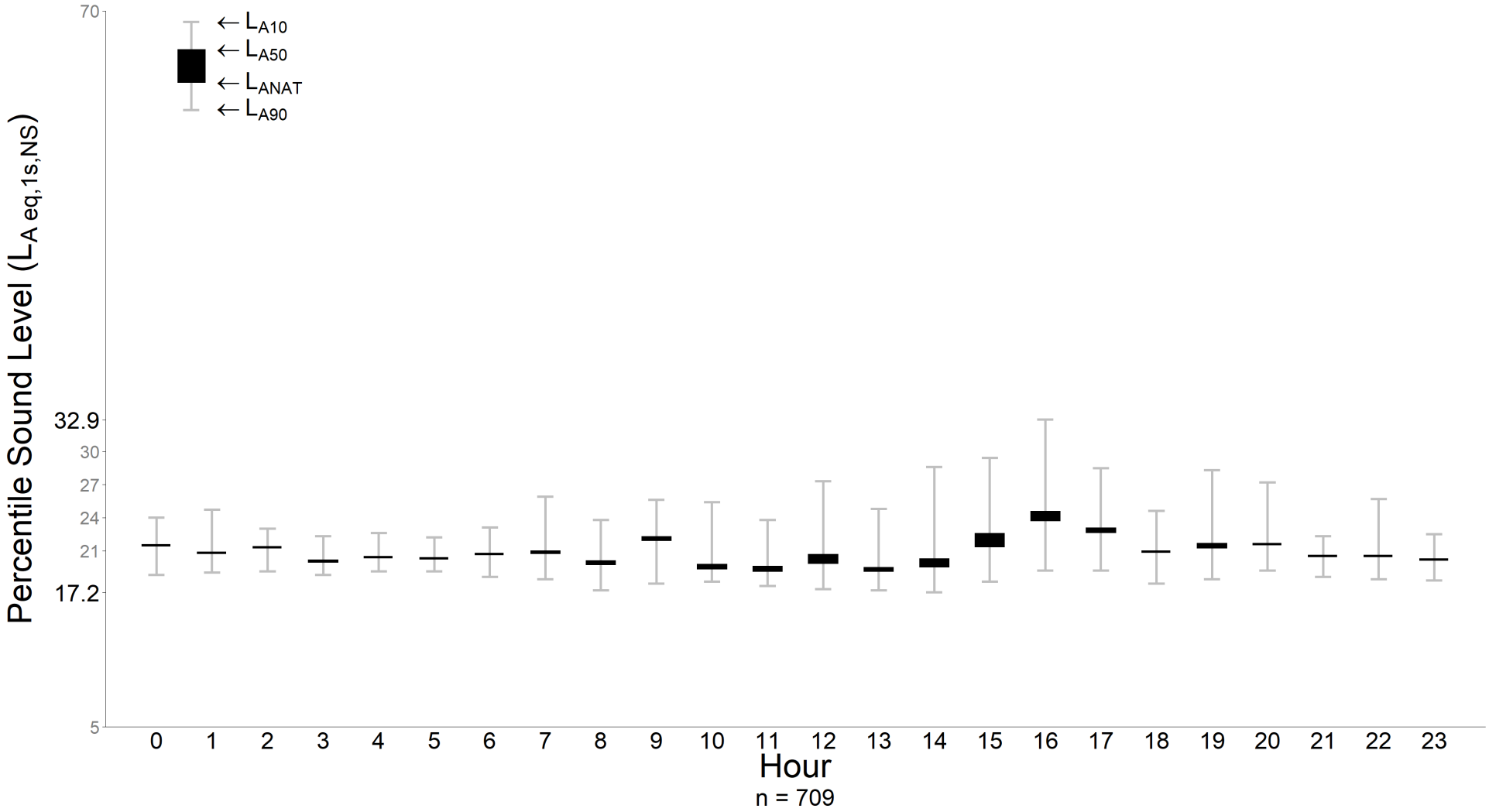


Figure 17. GAARFLOR: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

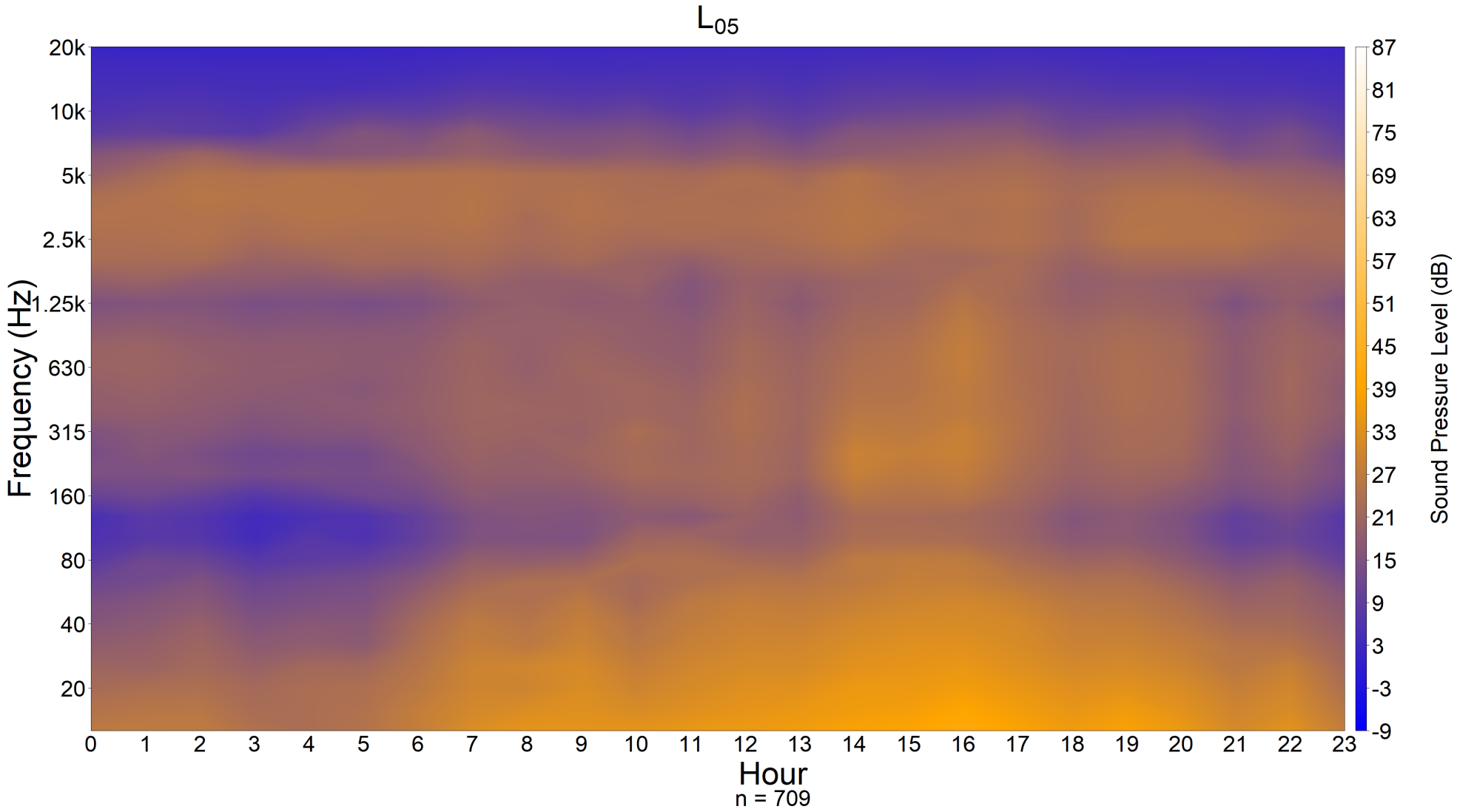


Figure 18. GAARFLOR: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

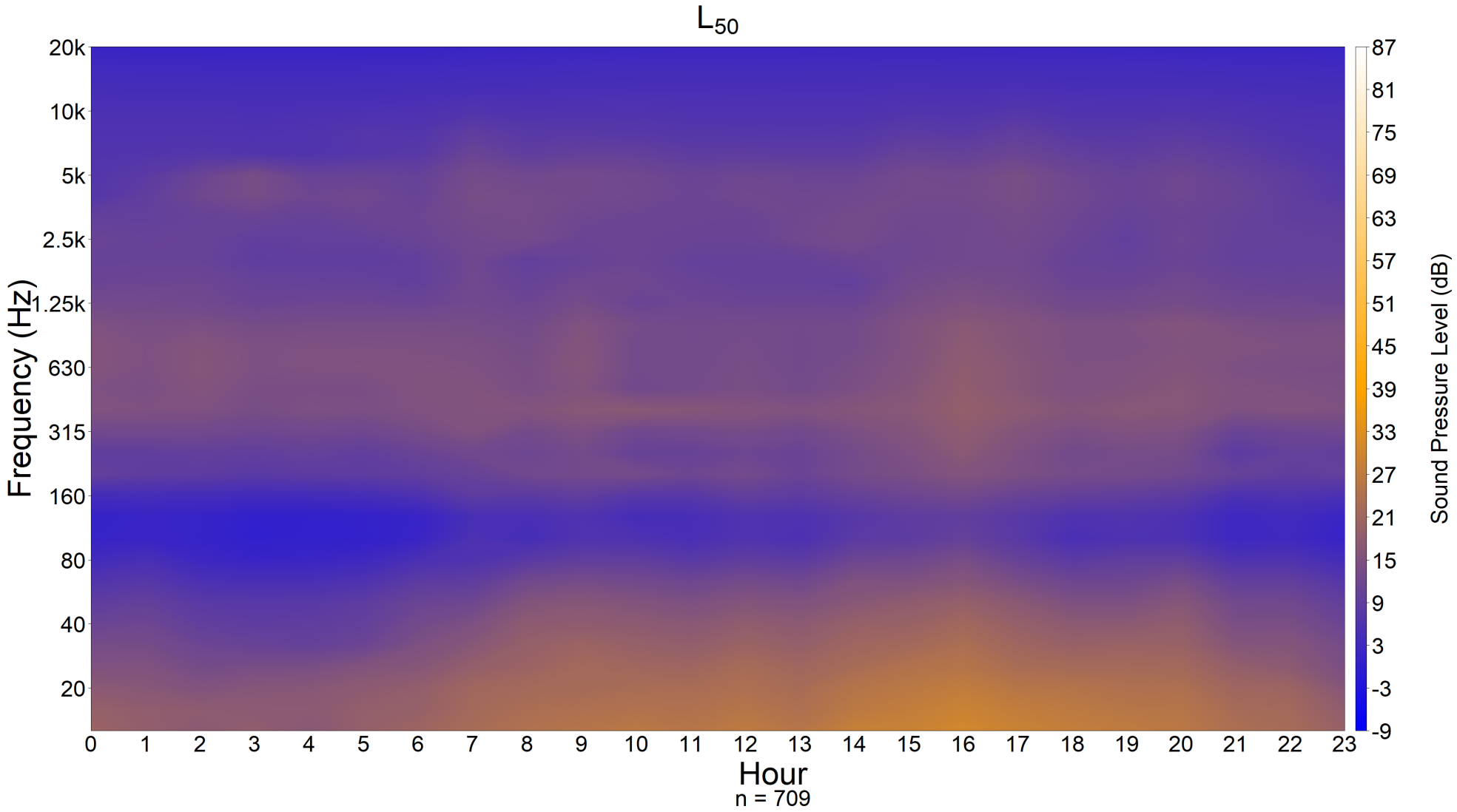


Figure 19. GAARFLOR: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

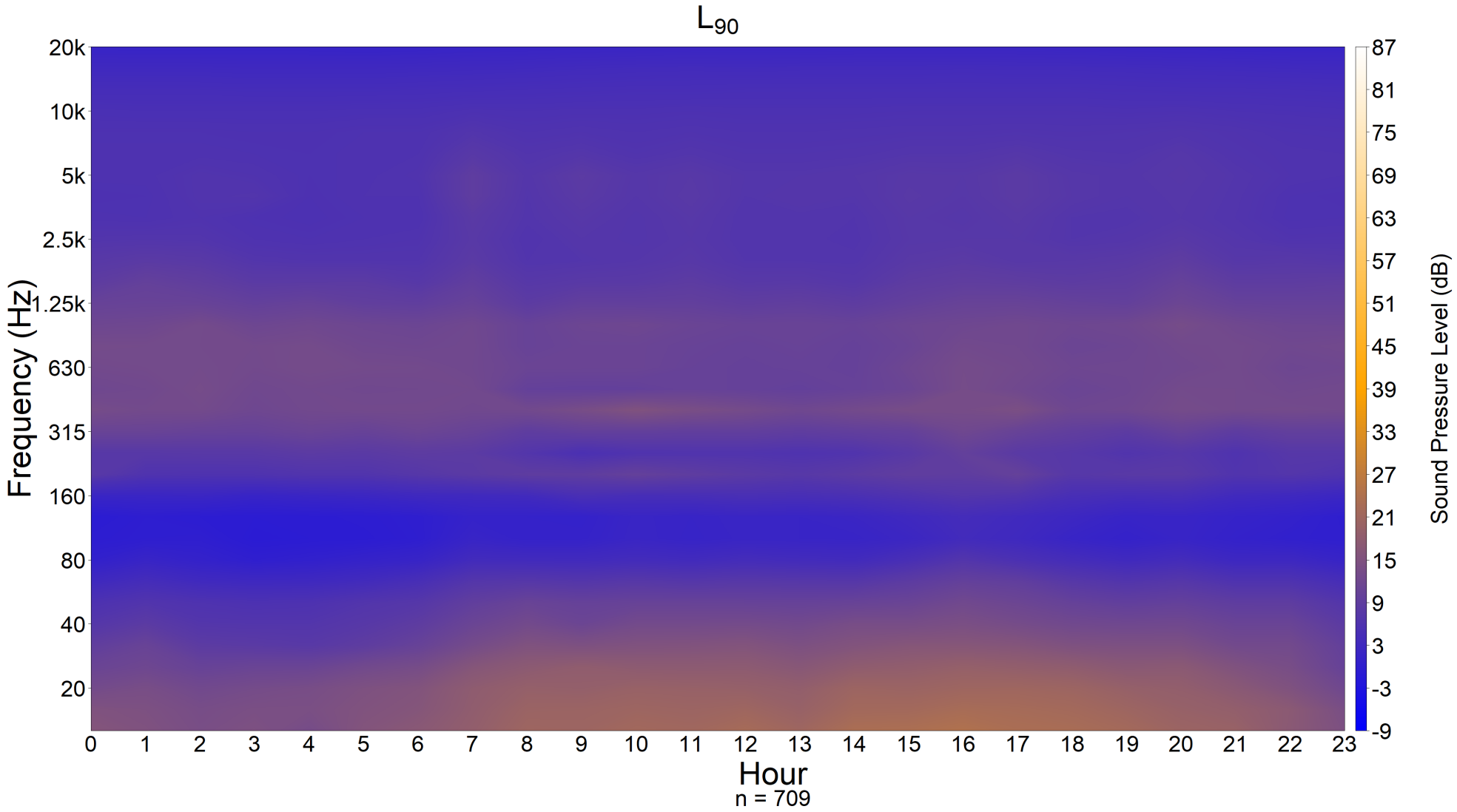


Figure 20. GAARFLOR: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

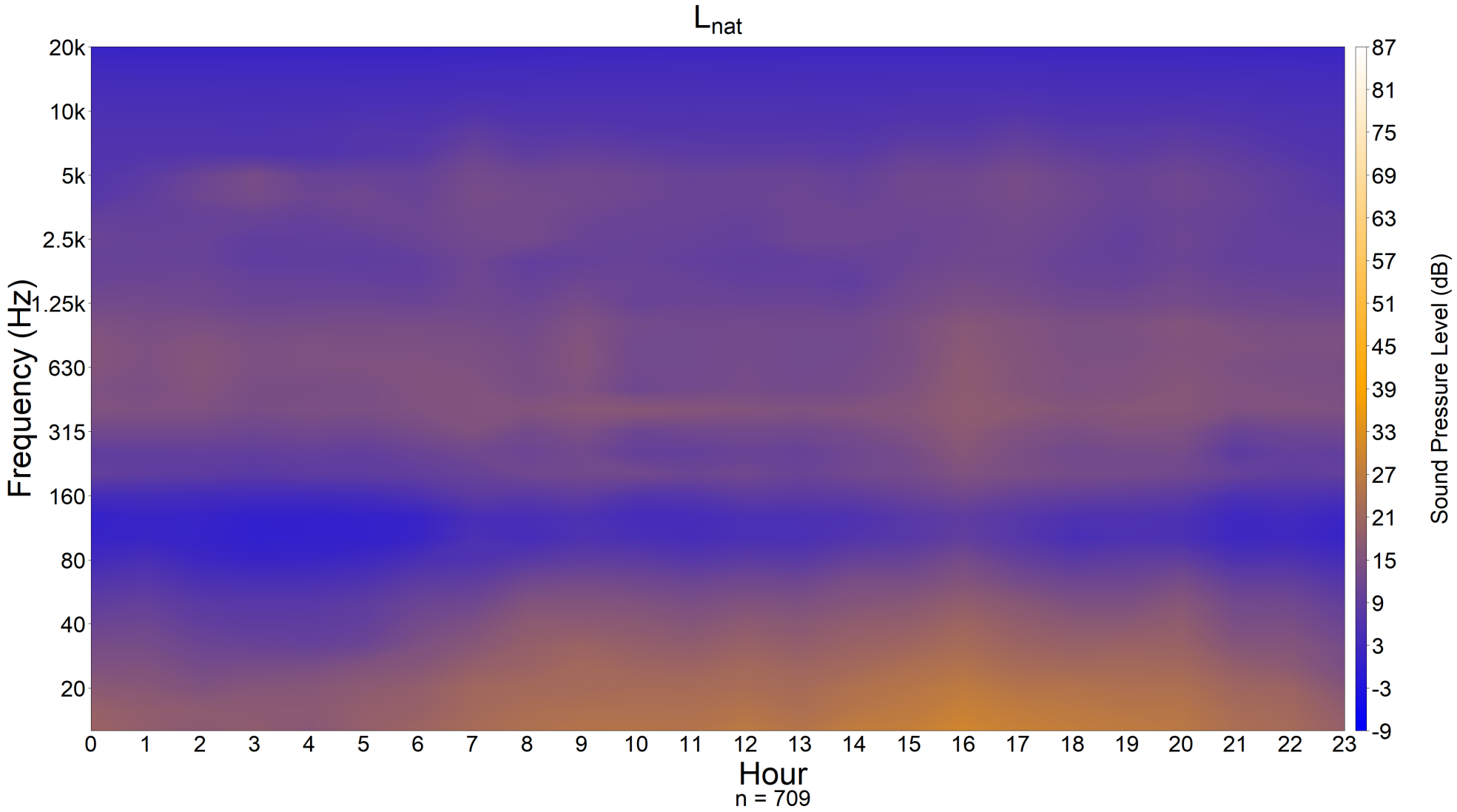


Figure 21. GAARFLOR: median and mean duration of noise−free intervals by hour of day, in Summer 2018.

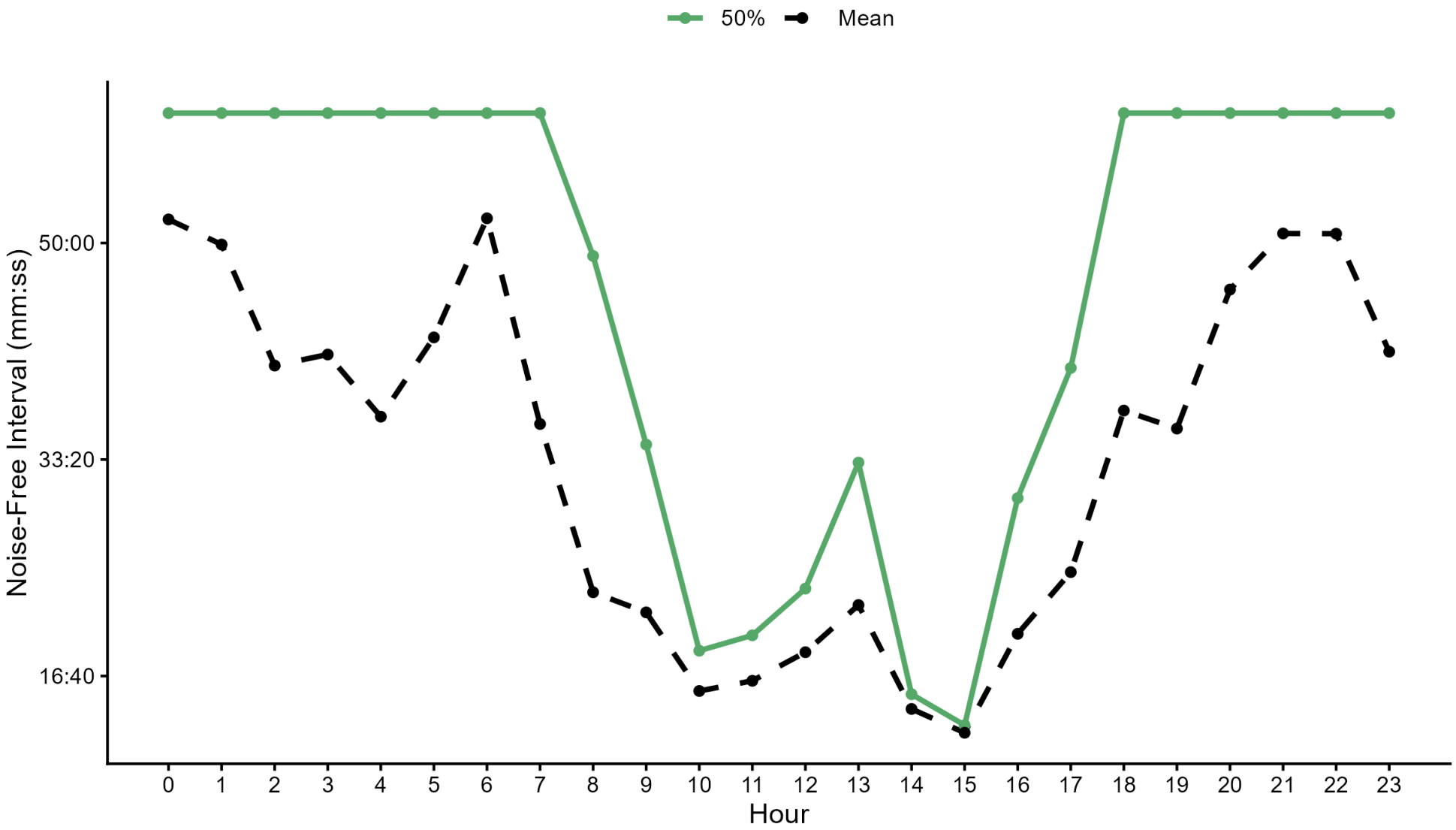


Figure 22. GAARFLOR: percent of time audible by sound source category (noise/natural), in Summer 2018.

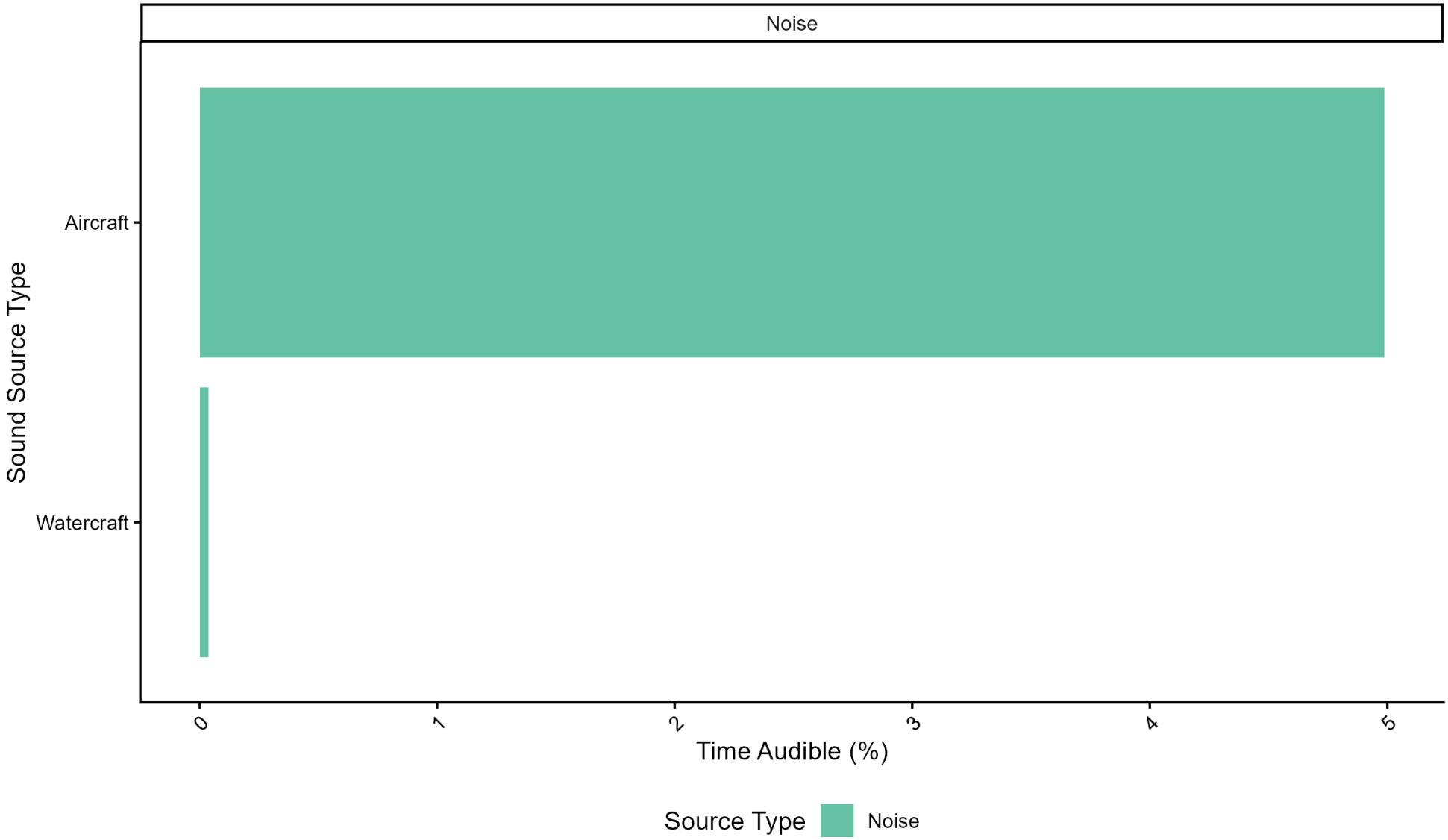


Figure 23. GAARFLOR: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2018.

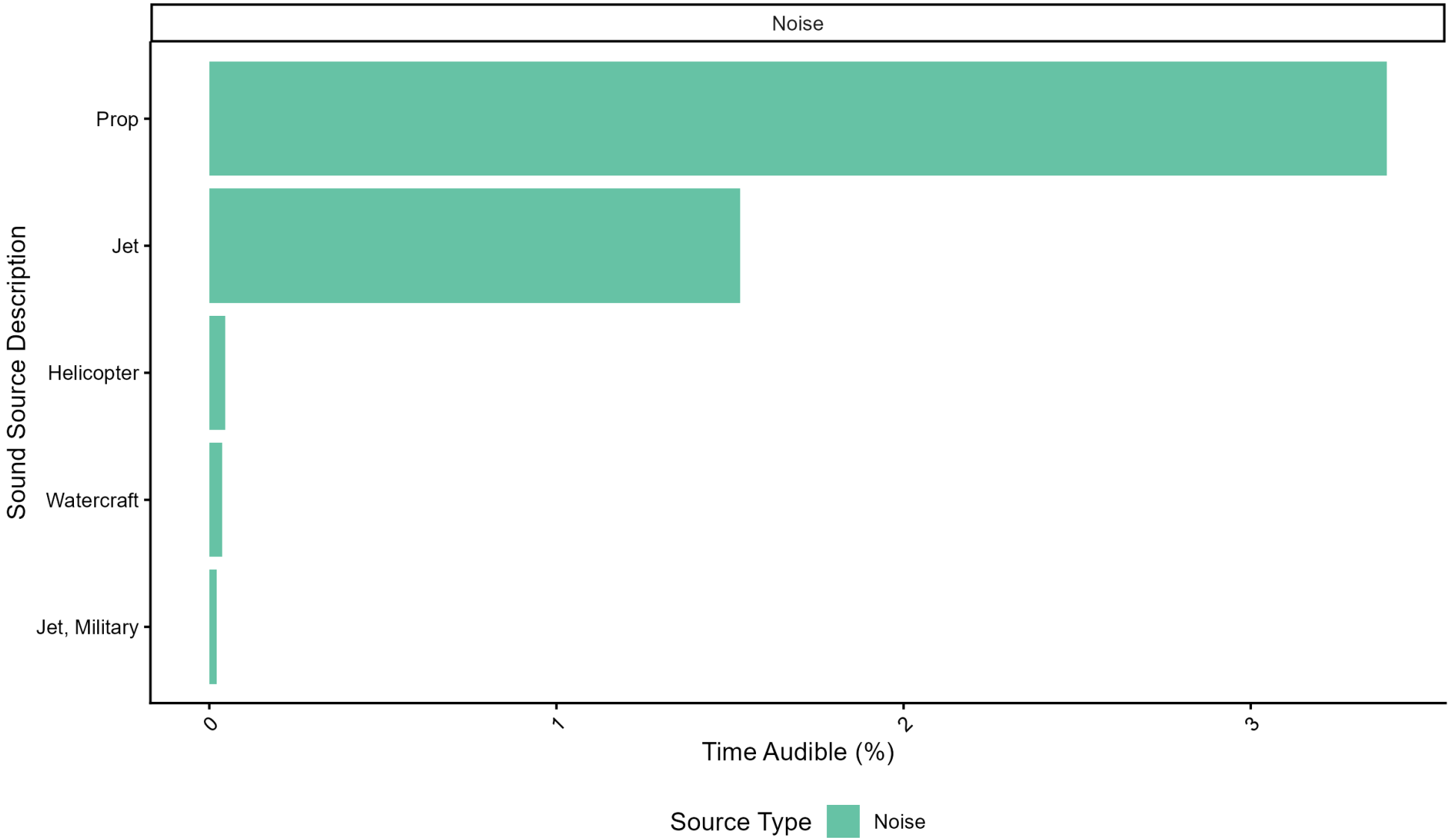


Figure 24. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARFLOR in Summer, 2018. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

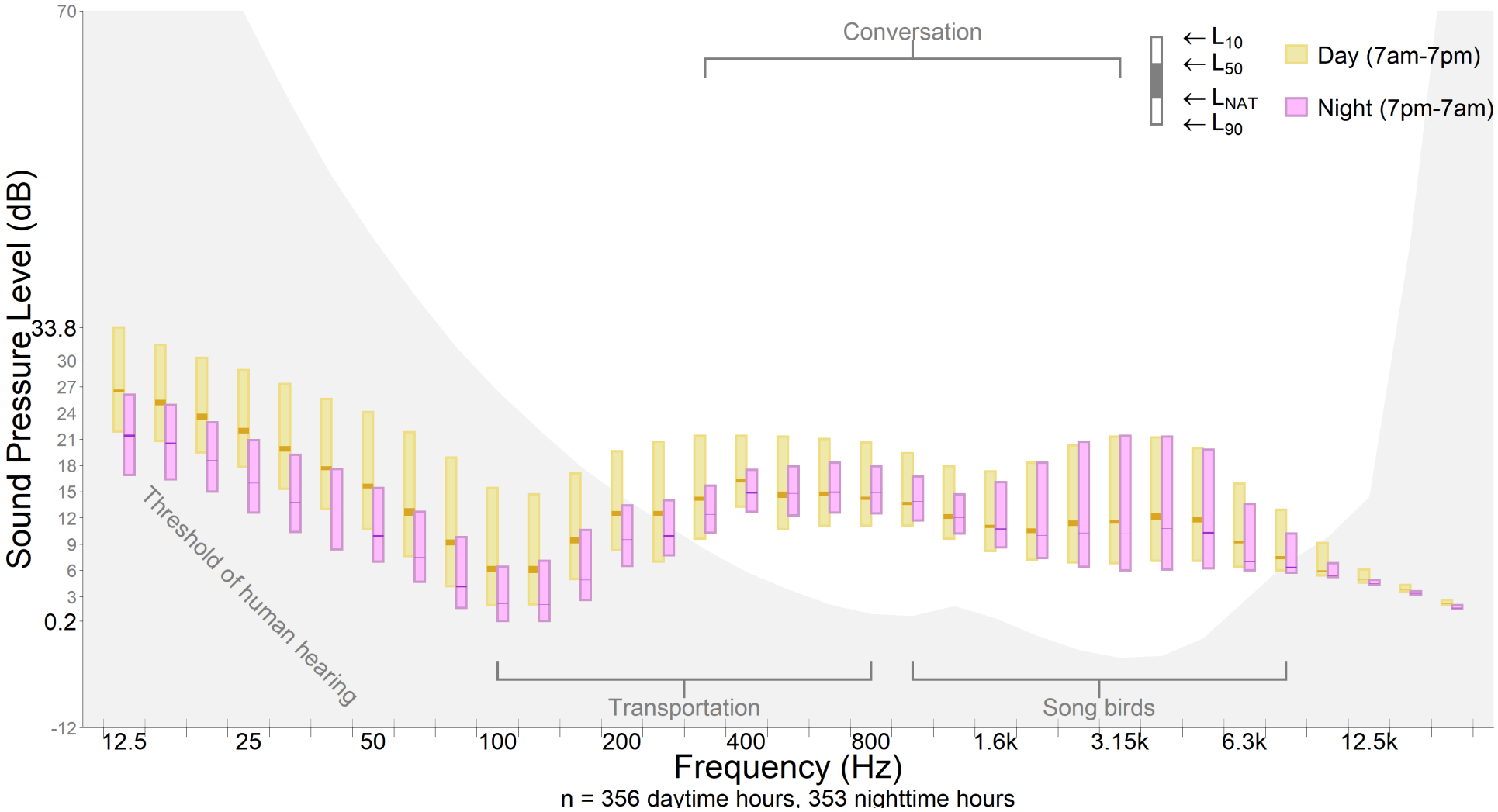


Figure 25. GAARUPNO: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2018.

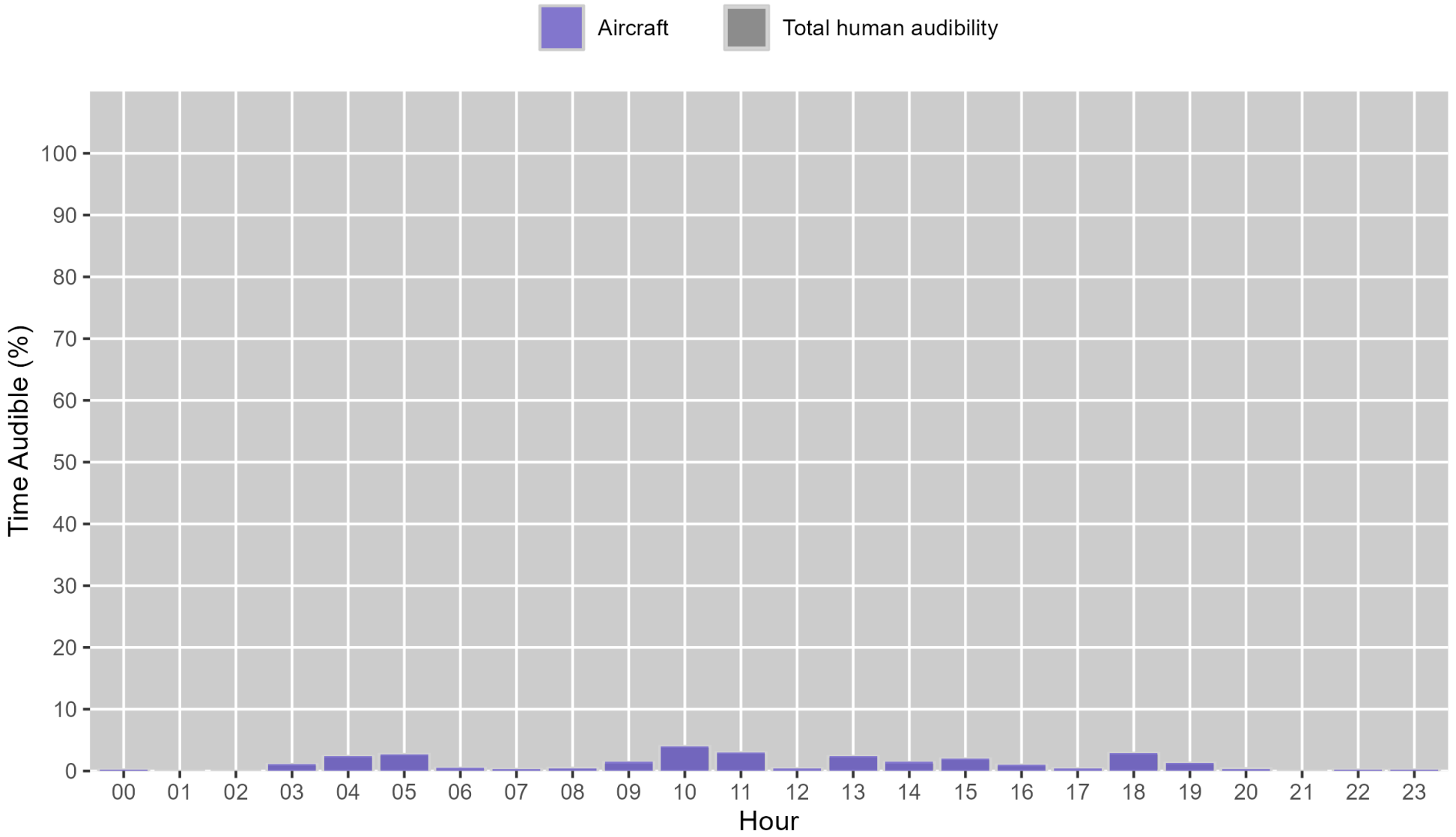


Figure 26. GAARUPNO: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2018.

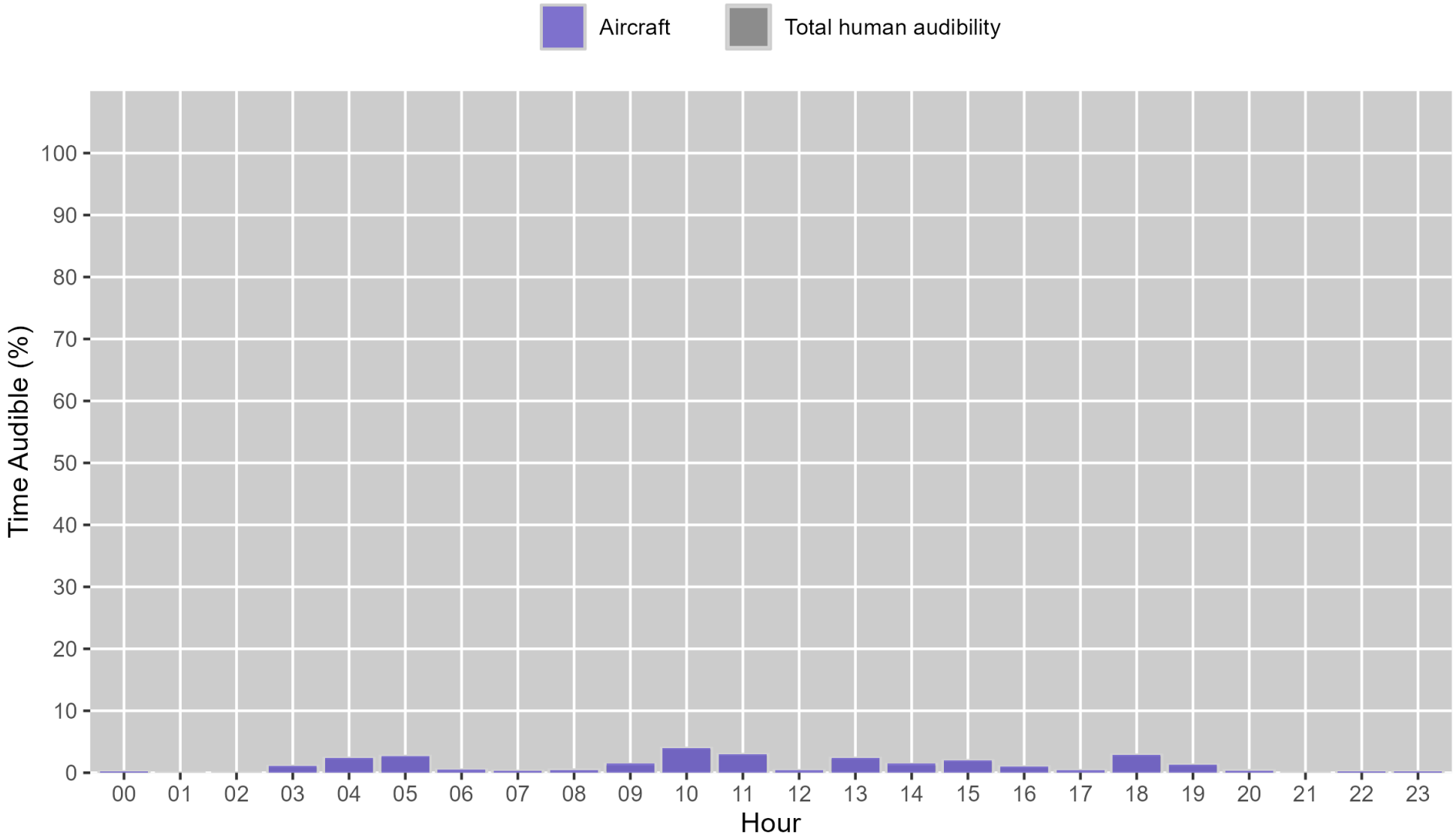


Figure 27. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARUPNO in Summer, 2018, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

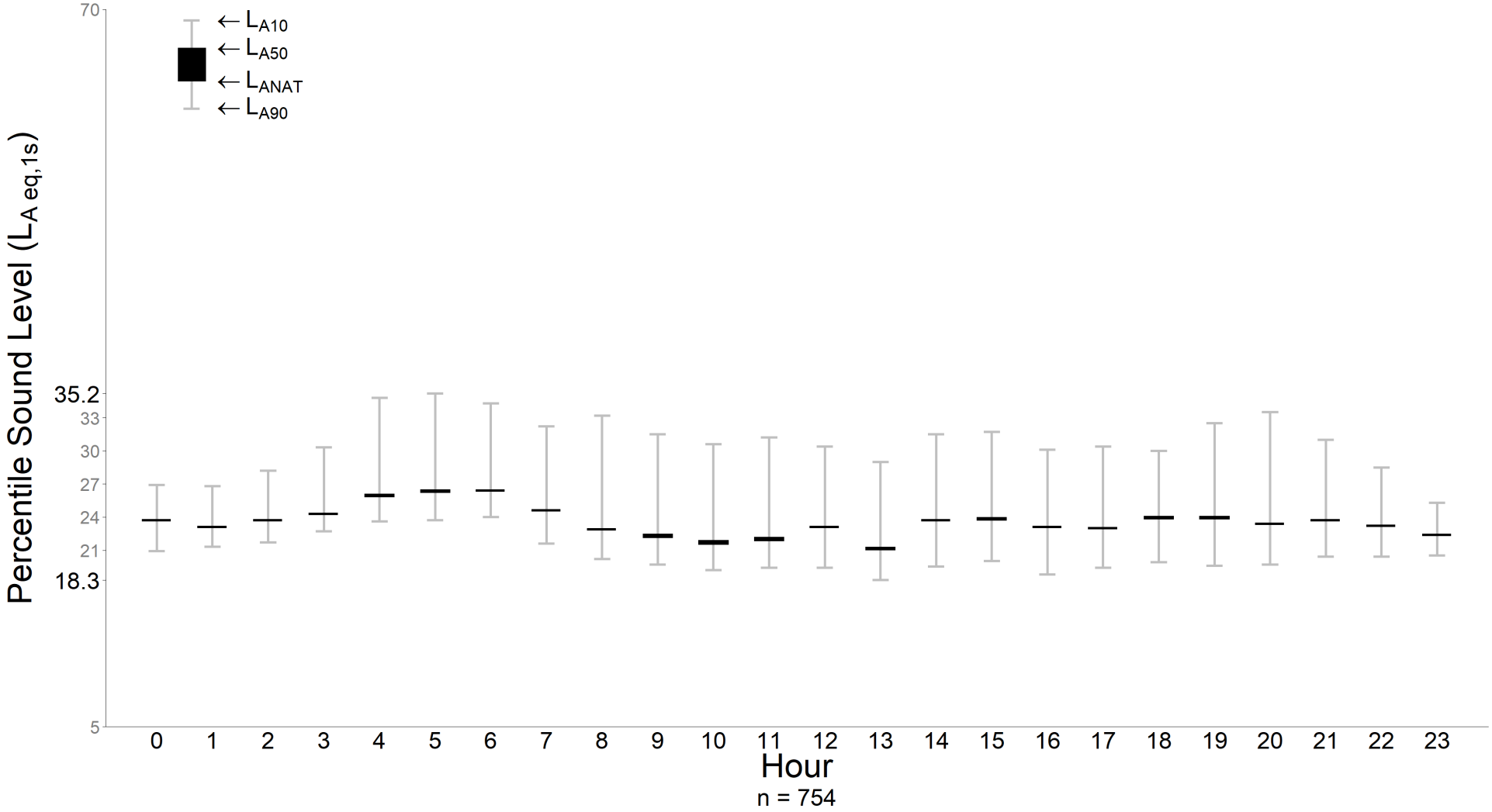


Figure 28. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARUPNO in Summer, 2018.

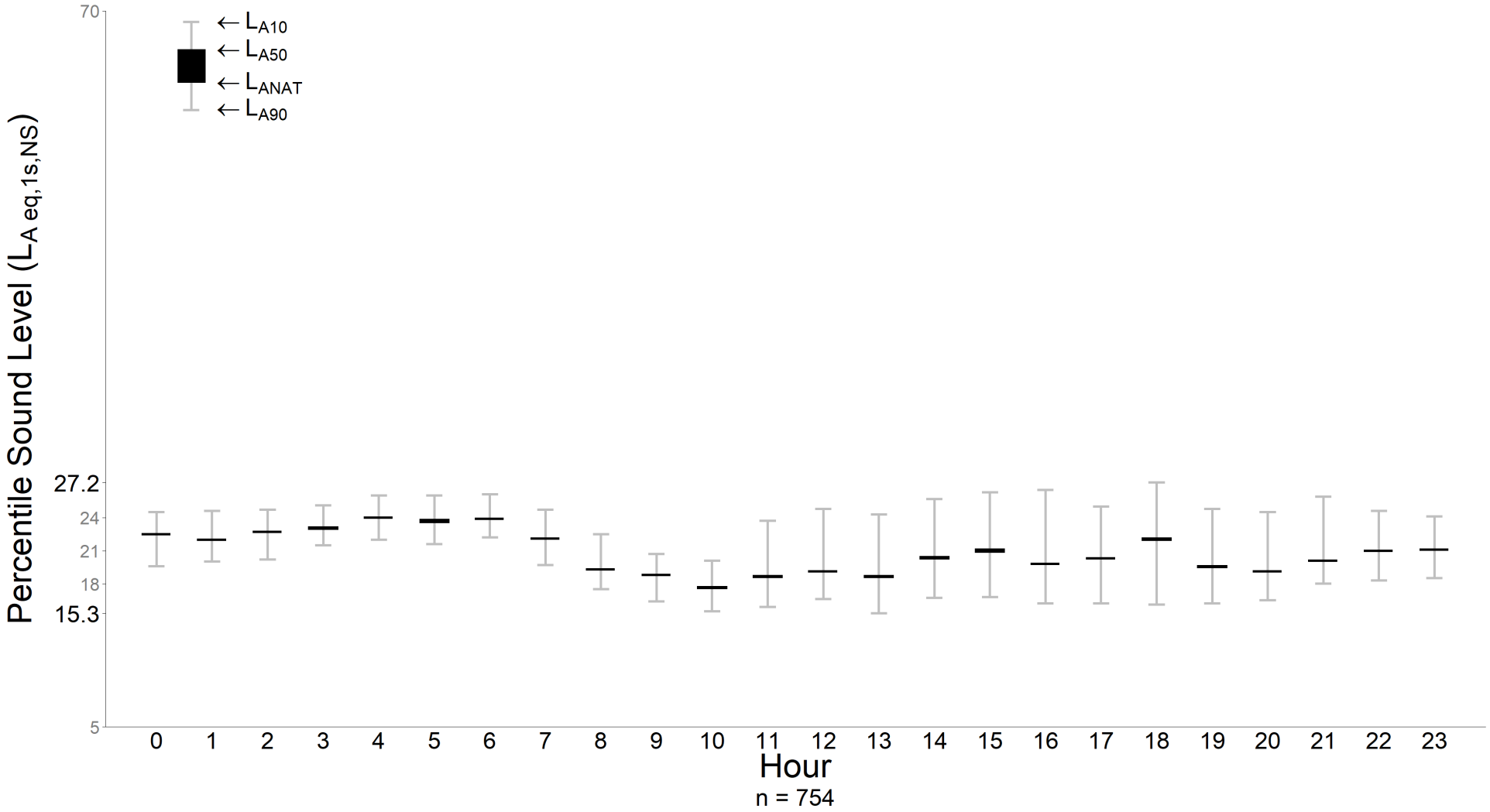


Figure 29. GAARUPNO: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

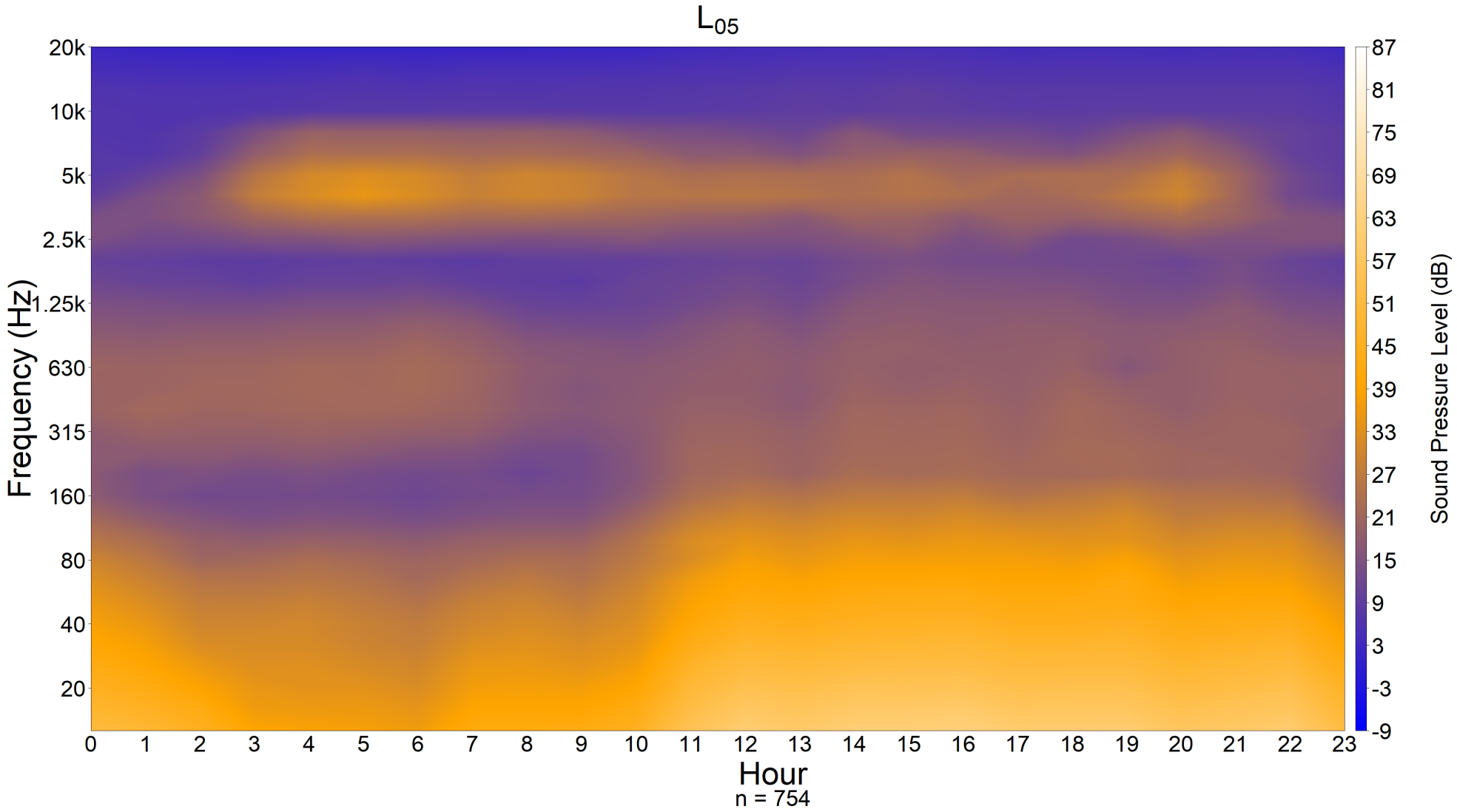


Figure 30. GAARUPNO: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

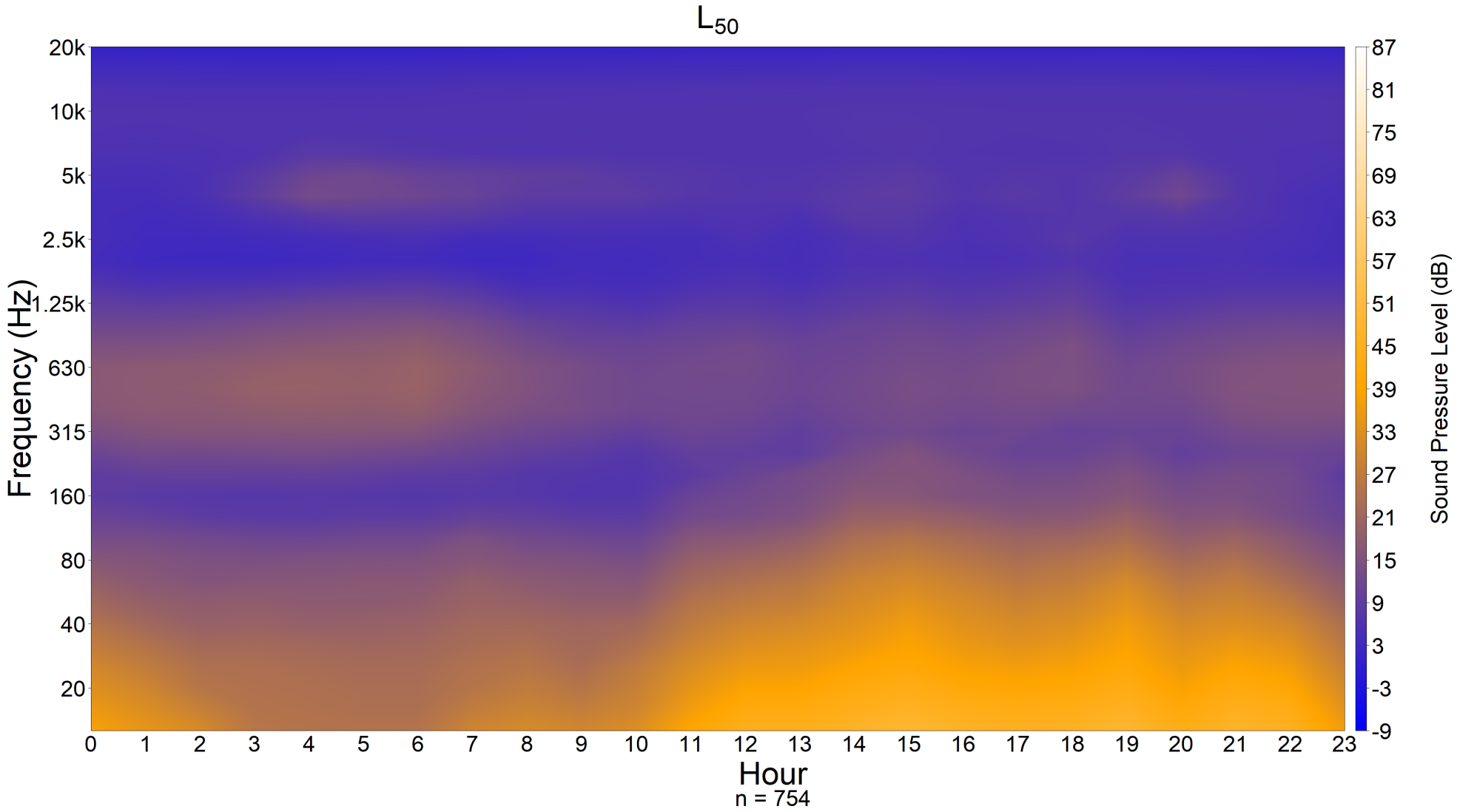


Figure 31. GAARUPNO: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

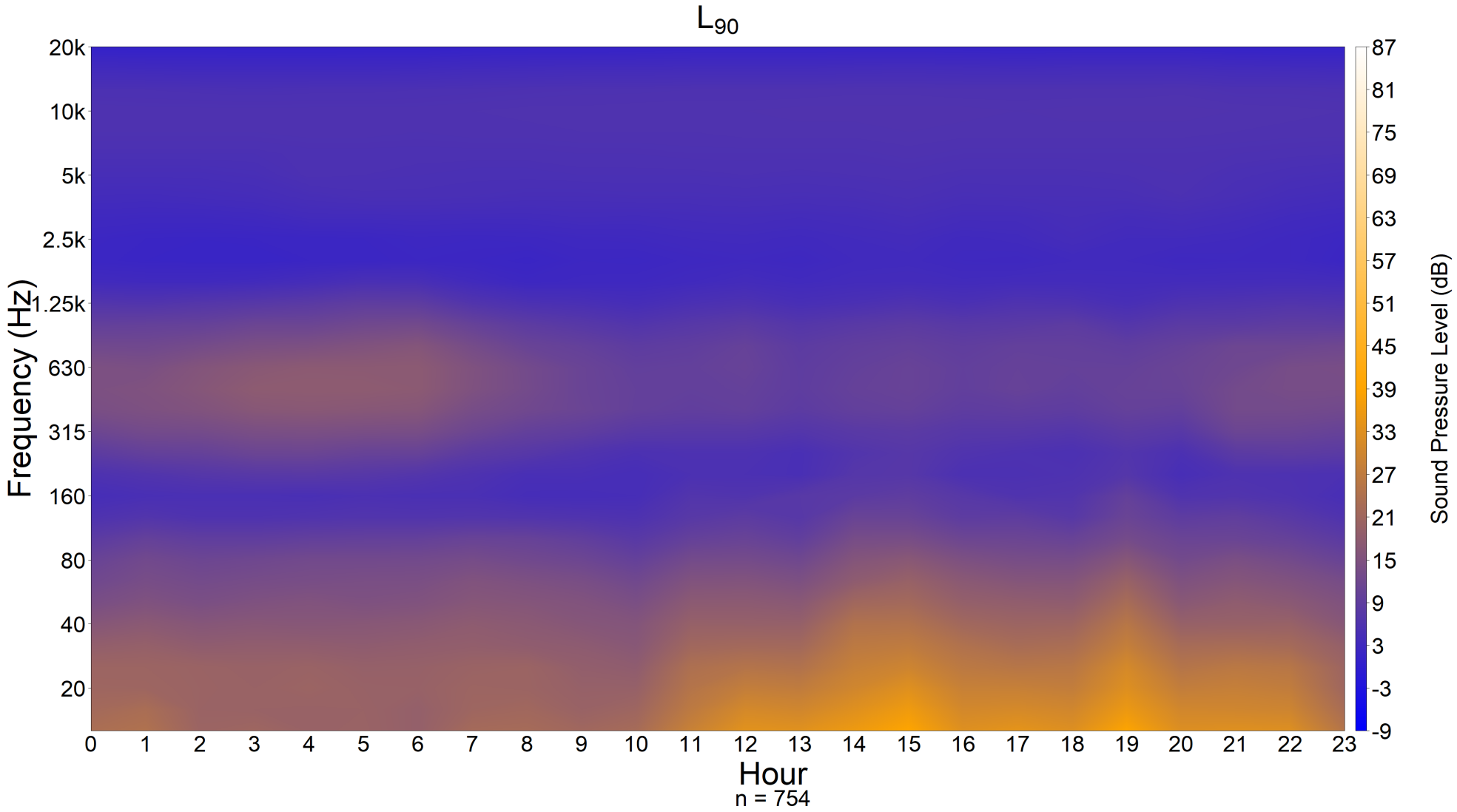


Figure 32. GAARUPNO: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2018.

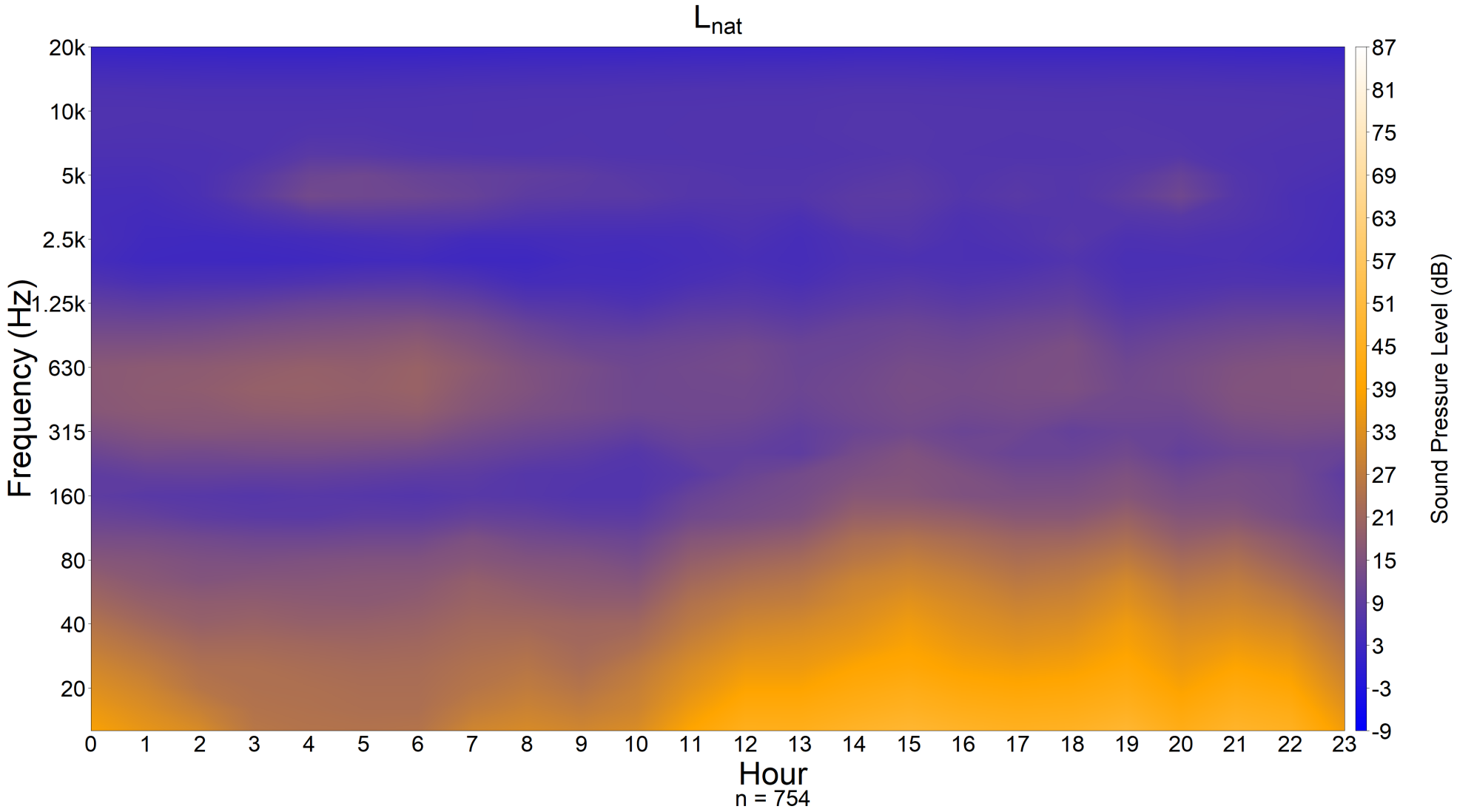


Figure 33. GAARUPNO: median and mean duration of noise−free intervals by hour of day, in Summer 2018.

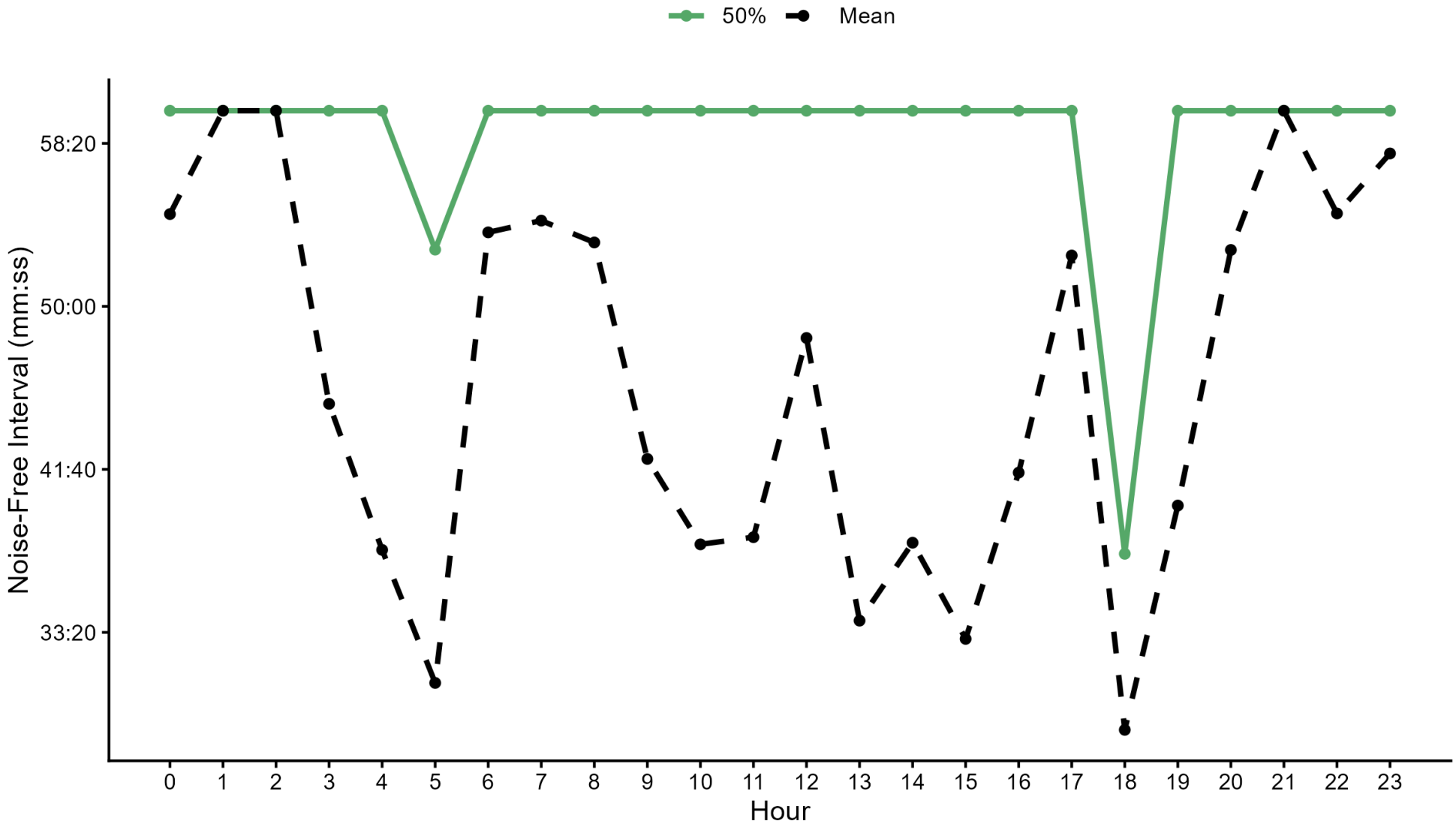


Figure 34. GAARUPNO: percent of time audible by sound source category (noise/natural), in Summer 2018.

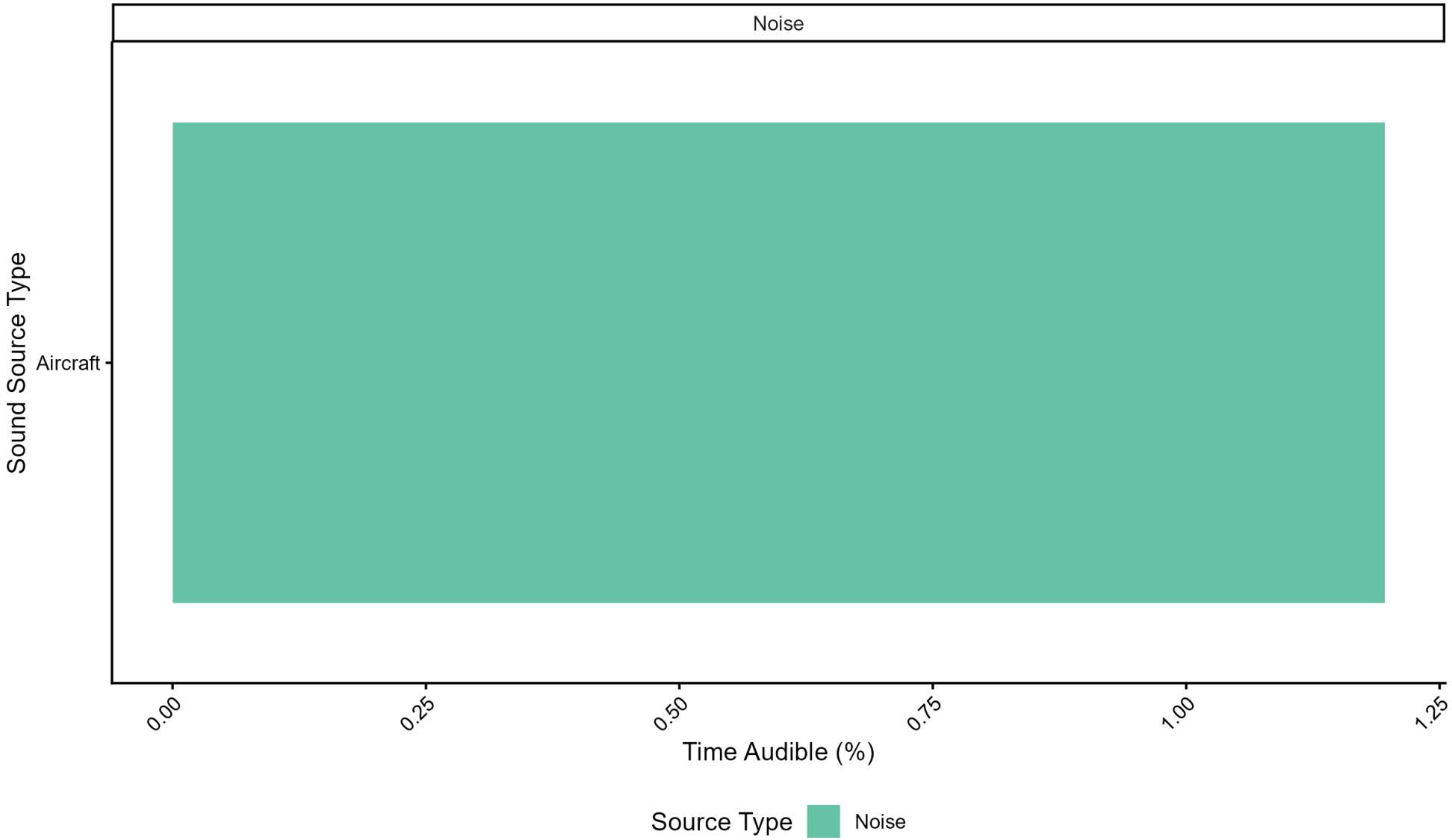


Figure 35. GAARUPNO: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2018.

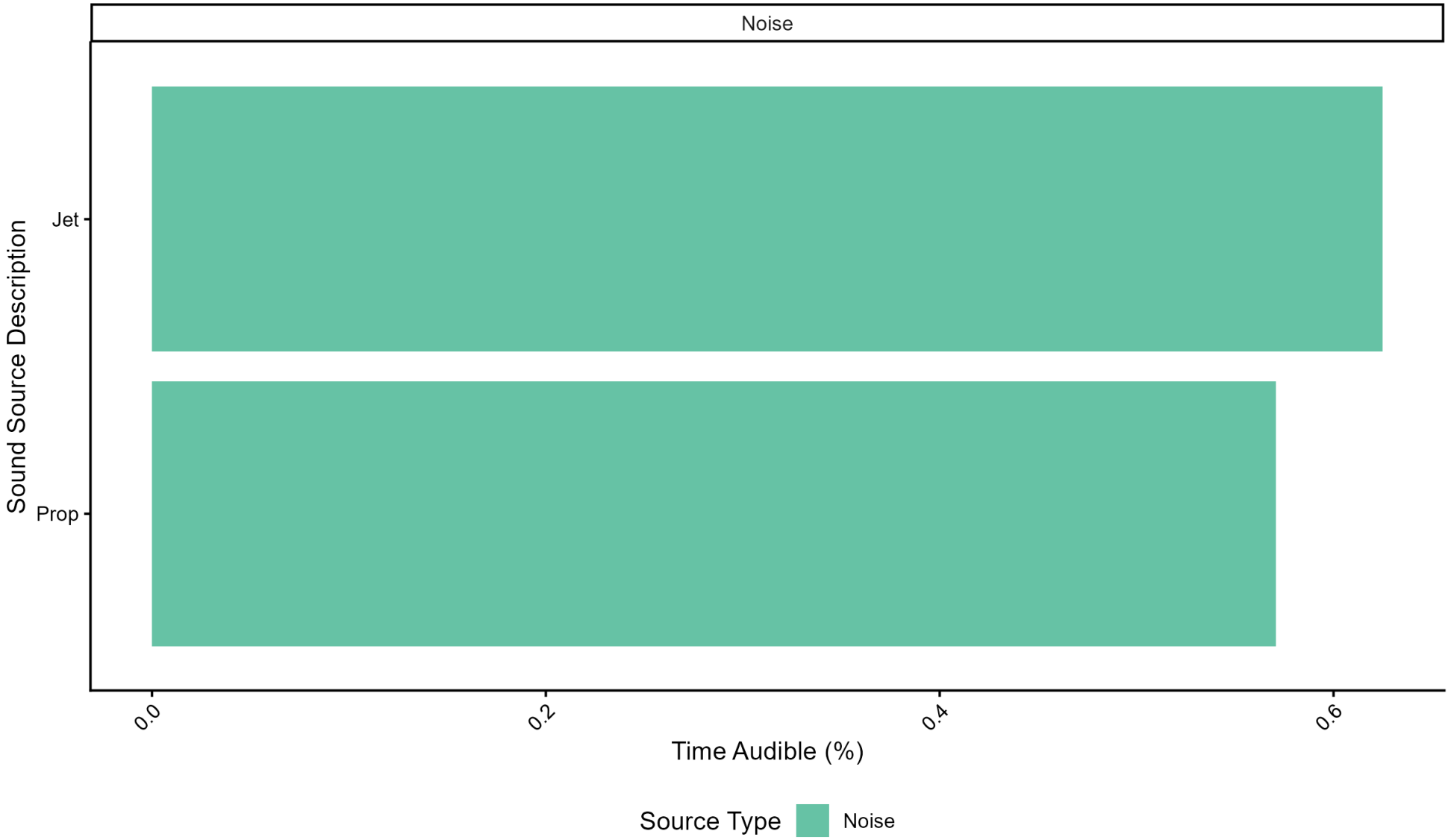


Figure 36. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARUPNO in Summer, 2018. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

